BMJ Open

BMJ Open is committed to open peer review. As part of this commitment we make the peer review history of every article we publish publicly available.

When an article is published we post the peer reviewers' comments and the authors' responses online. We also post the versions of the paper that were used during peer review. These are the versions that the peer review comments apply to.

The versions of the paper that follow are the versions that were submitted during the peer review process. They are not the versions of record or the final published versions. They should not be cited or distributed as the published version of this manuscript.

BMJ Open is an open access journal and the full, final, typeset and author-corrected version of record of the manuscript is available on our site with no access controls, subscription charges or pay-per-view fees (http://bmjopen.bmj.com).

If you have any questions on BMJ Open's open peer review process please email info.bmjopen@bmj.com

BMJ Open

How do electronic risk assessment tools affect the communication and understanding of diagnostic uncertainty in the primary care consultation? A systematic review and thematic synthesis

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-060101
Article Type:	Original research
Date Submitted by the Author:	20-Dec-2021
Complete List of Authors:	Burns, Alex; University of Exeter, College of Medicine and Health; Three Spires Medical Practice, Donnelly, Brian; University of Exeter Medical School Feyi-Waboso, Joshua; University of Exeter Medical School Shephard, Elizabeth; University of Exeter, Calitri, Raff; University of Exeter Medical School, Primary Care Tarrant, Mark; University of Exeter Medical School, Medical School Dean, Sarah; PenCLAHRC University of Exeter Medical School,
Keywords:	PRIMARY CARE, QUALITATIVE RESEARCH, Risk management < HEALTH SERVICES ADMINISTRATION & MANAGEMENT

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

How do electronic risk assessment tools affect the communication and understanding of diagnostic uncertainty in the primary care consultation? A systematic review and thematic synthesis

Dr Alex Burns MBBS, MGHP, BA(cantab). GP Partner, Threespires Medical Practice, Truro. PhD Student, University of Exeter, College of Medicine and Health. ORCID 0000-0002-9931-9198

Brian Donnelly. Medical Student, University of Exeter Medical School

Joshua Feyi-Waboso. Medical Student, University of Exeter Medical School. ORCID 0000-0002-5417-4298

Dr Elizabeth Shephard PhD, MSc, BSc. Research Fellow, University of Exeter, College of Medicine and Health. ORCID 0000-0002-3610-3680

Dr Raff Calitri PhD, MSc, BSc. Research Fellow, University of Exeter, College of Medicine and Health. ORCID 0000-0003-0889-4670

Dr Mark Tarrant PhD, BA(hons). Associate Professor, University of Exeter, College of Medicine and Health. ORCID 0000-0002-7194-1428

Professor Sarah Dean, PhD, MSc, BSc. Professor, University of Exeter, College of Medicine and Health. ORCID 0000-0002-3682-5149

corresponding author: Dr Alex Burns, ab1265@exeter.ac.uk

For peer review only

How do electronic risk assessment tools affect the communication and understanding of diagnostic uncertainty in the primary care consultation? A systematic review and thematic synthesis

Alex Burns, Brian Donnelly, Joshua Feyi-Waboso, Elizabeth Shephard, Raff Calitri, Mark Tarrant and Sarah Dean

Abstract

Objectives: To conduct a systematic review and synthesise qualitative research of electronic risk assessment tools (eRATs) in primary care, examining how they affect the communication and understanding of diagnostic risk and uncertainty. eRATs are computer-based algorithms designed to help clinicians avoid missing important diagnoses, pick up possible symptoms early and facilitate shared decision making.

Design: Systematic search, using predefined criteria of the published literature and synthesis of the qualitative data, using Thematic Synthesis. Database searches on 27th November 2019 were of MEDLINE, EMBASE, CINAHL and Web of Science, and a secondary search of the references of included articles. Included studies were those involving electronic risk assessment or decision support, pertaining to diagnosis in primary care, where qualitative data was presented. Non-empirical studies and non-English language studies were excluded. 5971 unique studies were identified of which 441 underwent full text review. 26 studies were included for data extraction. A further 2 were found from citation searches. Quality appraisal was via the CASP tool. Data extraction was via line by line coding. A Thematic Synthesis was performed.

Setting: Primary Care

Results: Electronic risk assessment tools included differential diagnosis suggestion tools, tools which produce a future risk of disease development or recurrence, or calculate a risk of current undiagnosed disease. Analytical themes were developed to describe separate aspects of the clinical consultation where risk and uncertainty is both central and altered via the use of an eRAT: “Novel risk”, “Risk refinement”, “Autonomy”,” Communication”, “Fear” and “Mistrust”.

Conclusion: Despite little evidence on the effectiveness of electronic risk assessment tools in primary care, their role is likely to increase. Understanding how they influence consultations may help both clinicians and patients in their interaction with eRATs.

Registered review with Prospero, CRD219446

Strengths and Limitations:

- This is the first review to address the impact of eRATs in the primary care consultation
- There is inconsistency in nomenclature surrounding these tools
- There is a lack of data from patients in the primary studies
- There are likely biases in the original studies.

For peer review only

Introduction

Diagnosis in primary care is complex and challenging. The main input, a patient's history and examination, can be variable, imperfect and subjective. Disease prevalence is lower than other health-care settings[1,2]. GPs are able to process and communicate risk and accept uncertainty in diagnosis:[3] Doing this in a way which allows a shared understanding of medicine with a patient is one of the seminal challenges of primary care.[4]

Clinical Decision Support Systems are designed to "enhance medical decisions with targeted clinical knowledge, patient information, and other health information"[5]. "Electronic Risk Assessment Tools" (hereafter referred to as eRATs) are decision support systems which help with diagnosis. They are software where information about a patient is processed electronically to produce an output pertinent to a diagnosis, either immediately or in the future. The output can be a quantitative risk of a specific diagnosis, or a more qualitative output: to consider a particular diagnosis or list of possible diagnoses.

eRATs can be considered as the computerised evolution of clinical scoring systems. Clinical scoring systems are "algorithms designed to predict outcomes, aid decision making, support treatment options, manage clinical risk, or improve efficiency". [6] Examples include the Wells[7] and the NEWS2 scores[8]. The use of a computer permits more complex algorithms and improves accuracy and speed of calculation. eRATs can enable extraction of data previously inputted into primary care IT systems, and can prompt users to input missing data. They can incorporate "pop ups" during a consultation to avoid missing important diagnoses.[9,10] They have a range of stated aims: making diagnosis more evidence based and objective; avoiding missed diagnoses; or facilitating shared decision making.

eRATs tend to target diagnoses which are both serious and benefit from early intervention,[9,10] an obvious example being cancer. To achieve adequate sensitivity these algorithms tend towards a low positive predictive value.[9,10] As eRATs use more complex calculations, invisible to the clinician, and insert themselves into the clinician consultation more proactively, they may come to represent a "third actor" in the consultation, alongside the patient and clinician.[11] They have the potential to change the consultation dynamic and diagnostic thinking process. Although there have been systematic reviews conducted on the effectiveness of eRATs[12,13], There have been no systematic reviews using qualitative methods examining risk and uncertainty in their use.

Aim

This study aims to synthesise qualitative research on eRATs when used making diagnoses in the primary care consultation, examining how they affect the communication and understanding of diagnostic risk and uncertainty.

Methods

Study Selection

Searches of MEDLINE, EMBASE, CINAHL and Web of Science were performed. The search strategy[14] targeted studies involving risk assessment or decision support tools for diagnosis, delivered by electronic or computerised means, within a primary care setting, where qualitative data are

For peer review only

reported regarding risk or uncertainty. Non-empirical studies and non-English language studies were excluded, as well as studies outside of primary care, or where qualitative data was not presented. Using Covidence software[15], three authors (AB, JF-W and BD) independently screened titles and abstracts, and then full text for relevance. Disputes were resolved on discussion between authors. The initial 5971 unique studies that were identified reduced to 441 on title/abstract review. These underwent full text review, which yielded 26 individual studies, and one systematic review[12] (figure 1). These were hand searched for further citations, which yielded 2 additional studies. Key eRATs embedded into System1, a mainstream UK IT system, were identified and the developers emailed (where qualitative studies had not already been identified) to see if there were unpublished data available. This did not identify further studies. Thus 28 independent studies were included for data extraction (supplementary 1)

Quality appraisal Data Extraction

The 28 studies were assessed for quality (AB, ES and SD) using the CASP critical appraisal checklist[16]. Discrepancies were resolved on discussion between the authors (supplementary 2). None were excluded from analysis due to low quality[17]. Data extraction was performed by AB. Quotes and researcher statements regarding risk and uncertainty were extracted electronically and coded into NVivo software.[18]

Thematic Synthesis

Thematic Synthesis was chosen as it is based in Scientific Realism[19]: the purpose of this review is to examine how clinicians and patients interact with eRATs. For the results to be broadly applicable, the underlying assumption is that this interaction is consistent, or consistent within a range.

Descriptive codes were developed by AB. Codes were described in the initial studies coded, and subsequent studies coded into these concepts, with new descriptive codes being created, refined or ordered hierarchically when deemed necessary. These descriptive codes were discussed with other authors (SD, ES, RC and MT). Thematic Synthesis involves the "translation of concepts from one study to another"[20]. The ten studies with analytical findings pertaining to risk or uncertainty were examined. The developed descriptive codes were applied to each to explore new coding approaches. The descriptive codes were then developed inductively into analytical themes describing separate aspects of the clinical consultation where risk and uncertainty is both central to, and altered via, the use of an eRAT. Selective quotes are used to illustrate the richness of data and to provide support for these themes.

Patient and public involvement

Patients and the public were not involved in the review

Research ethics approval

This study did not receive or require ethics approval, as it does not involve human or animal participants

Results

Included studies were examined using methods borrowed from Meta-study[21]. It became evident that there were three conceptually different forms of eRATs: those which produce a risk of an immediate diagnosis; those which produce a risk of a future diagnosis; and those which produce a list of potential possible diagnoses. It was decided to synthesise data across these groups, to explore the themes across these eRATs, to see if there were findings consistent between them[22], acknowledging the aspects of the different eRATs which are the same and contributory to the aims of the review.

Included studies had publication dates ranging from 2003 to 2019. The target diagnosis was most frequently "Cancer", with six targeting a selection of cancers,[23–28] four breast cancer,[29–32] and one colorectal cancer.[33] Four targeted cardiovascular risk[34–37] and four secondary prevention of stroke,[38–41] with a further two addressing risks of both cancer and cardiovascular disease[42,43]. One targeted diabetic complications,[44] one diabetic foot disease,[45] one mental health risk,[46] and another pharyngitis or cough.[47] One study asked clinicians about their experience with any eRAT[48]. Two examined the use of differential diagnosis generators.[49,50]

Data collection was principally via focus groups and interviews, but a "think aloud" process[51] was documented as well. Most studies focused on clinicians,[23–26,28,29,33,37–39,42,43,45,48] and thirteen collected data from patients, either exclusively[30,32,35,36,40,44] or together with clinician data.[27,31,41,46,47,49,50] Use of an eRAT with an unfiltered patient population was restricted to thirteen of the studies.[25,26,28,32,35–37,41,43–45,47,50] Otherwise, simulated clinics were used [24,33,39,49] or a demonstration of the eRAT or a prototype.[23,27,29–31,36,38,40,42,46]

Analytic approaches adopted by the studies varied, as was the consistency in reporting them. Eleven studies reported use of Thematic Analysis,[23,27,28,32,35,40,41,45,46,49,50] three described a framework analysis using Normalisation Process Theory[24,26,33] and one used Grounded Theory.[48] Otherwise, analysis was described without giving it a methodological name. Only ten of the studies reported themes or findings pertaining to risk or uncertainty in their analysis.[28,32,35–38,40–42,44]

Analytical Themes:

Six analytical themes are presented below with supportive quotes (further supportive quotes are presented in supplementary 3).

"Novel risk". A new risk is introduced into the consultation, either as an unconsidered diagnosis or an unexpected high risk.[24,26,31,35,48,49] It brings the risk to the forefront of the consultation and ensures that it is adequately considered.

"It's just electronic highlight, makes you think even if you immediately dismiss it, at least that millisecond you've thought about it, and I think that is going to be useful at some point, but for how many people I don't know."[26]

The eRAT can raise the risk of alternative diagnoses:

"Yes, I must admit ovarian didn't come so high up, I had already fixated on the bowel stuff, which maybe I shouldn't have. This really said hey, consider ovarian as well. So again that was useful, yes."[24]

For Peer review only

"Risk refinement". The risk is processed into a more accurate assessment. It enables clinicians:

"to *take complex, competing information and bring it together".*[39]

This refinement of risk serves several purposes. It can give an objective single point estimate of risk.[31,39] it can take complex risk factors and combine them into a single risk output,[40] and it can orientate the clinician towards guidelines and clinical pathways, triggering them to add missing information to the eRAT,[37] suggesting the next diagnostic tests which are appropriate.[24]

eRATs can introduce clinicians to new knowledge or raise awareness of new or existing best practices.[26,40]

"Communication". eRATs assist clinicians to communicate risk to patients, and in patient-facing eRATs, to help patients communicate with their clinicians[27]. They can provide a focus point for a consultation.[37,48] Clinicians can use the eRAT output to justify their next actions: be this referral or further testing[24,47] or, where the risk output is low, inaction.[24,33]

"OK, so our little risk calculator here is recommending that we would swab you for strep throat, and I agree with that."[47]

"I think FRAX [is] useful because [it] can cut down on unnecessary investigations."[48]

The eRAT can emphasise the level of risk, with a preference for colour displays or other qualitative methods of showing risk.[37,48] eRAT outputs can be used as a motivator to take treatments or change behaviour.[41]

"Autonomy". The eRAT output is a personalised risk pertaining to a patient's health, enabling them to make informed decisions. This is regarded as of innate benefit.[31,40]

"Anything that gets people to have a think about their health is a good thing, anything at all."[33]

"Fear". The eRAT could create anxiety or disquiet in both patient and clinician. This can occur at several levels. The first is the fear of a serious diagnosis when the eRAT triggers. Patients have understandable concern about their own health and welfare:[36] clinicians share this fear, with an overlay of concern about poor understanding of risk among patients:

"And I thought, 'Wow! Patients coming with these tummy pains,' even if I'm thinking, 'She's got a 10% chance,' to actually think, 'Wow! Listen, you got a 38% chance of cancer.' General public are very—I would find that confronting and the general public have a shocking record at understanding risk."[24]

A second fear was of the eRAT interrupting how clinicians normally think about diagnosis.[49,50] The eRAT could distort how they handled diagnostic risk and unwanted risk suggestions made clinicians uncertain. There was a fear that the risk outputs could lead to a loss of control of the consultation, as the uncertainties raised by the eRATs override other agendas.[24,48]

Finally, there were fears expressed regarding the medico-legal consequences of eRATs. Clinicians expressed concern that any eRAT output, if not acted on, could act as evidence against them[26,50] and make them vulnerable to litigation. Here though, there was a negative case in the data, with one clinician reporting the eRAT was reassuring: that it acted as a "backstop", reducing fear of missing a diagnosis:

For peer review only

"So although it can cause irritation, the flip side of that is you can relax a little bit more and that you don't have to remember absolutely everything (…) You know, how to sort the wheat from the chaff. So although people might think it's irritating, it can be reassuring."[26]

"Mistrust". There was a lack of trust in the risk output of the eRAT. Some clinicians felt that the eRAT algorithm was flawed: because the trigger was set at an incorrect level of risk;[25,50] that it relied on a single parameter, and changes to that parameter could disproportionately change the output;[37] that the inputs themselves were out of date or flawed in some way; or that the outputs were outside the control of patient and clinician. Thus there was suspicion that the eRAT could be either under or over reporting a risk.

"Inevitably it will come up with a diagnosis of cancer. When the patient walks through the door . . . quite clearly, quickly, we can see this is not a serious diagnosis"[50]

There was a preference from both clinician and patient for diagnostic tests rather than the eRAT.[32,33] Here, a diagnostic test would be ordered despite the eRAT suggesting a risk low enough that testing was unnecessary: the test was trusted and the eRAT discounted.

There was faith in personal judgement over the eRAT. Patients preferentially trusting their knowledge of their own bodies,[31,44] clinicians trusting their clinical acumen or gut instinct.[24,33,42,50] Over-ruling an eRAT with clinical acumen was seen as good medicine:

"Patients don't follow flow charts, in my experience."[50]

Discussion

Summary: This review found that eRATs can be helpful in considering and communicating risk and uncertainty during diagnosis. The first four themes are broadly congruous with the stated aims of eRATs. "Novel risk" can reduce premature closure in diagnosis. Primary care diagnosis has a reliability issue,[52] and risks losing credibility with an increasingly risk-intolerant public if serious diagnoses are missed[1]. "Risk refinement" could lead to increased consistency of diagnostic decision making, which can improve adherence to guidelines, and play a part in cost containment.[5] The process of shared decision making between clinician and patient requires a combination of the review's themes of communication, autonomy and risk refinement.[53] The themes of "mistrust" and "fear" represent problems with the use of eRATs.

Strengths and limitations: This review is the first to address how eRATs affect perceptions of risk and uncertainty in the primary care consultation. The broad scope of this review means a range of eRATs have been synthesised. Some themes presented derive a greater part of their data from eRATs of a particular conceptual design, which could challenge their transferability across all eRATs. This review was intended to look at the wider clinician and patient interaction with eRATs and the findings are broadly transferable, which supports the analytical approach taken.

Another challenge concerns the nomenclature surrounding these tools: we used search terms to capture all studies where information was processed electronically to produce an output pertinent to a diagnosis, either immediately or in the future: here we have called them eRATs. The language used by the originators is often different and has changed over time. eRATs can also be referred to by other names now, such as Diagnostic Decision Support Systems, Electronic Screening Tools and

Risk Prediction Tools. The review may have missed interventions which, if examined, might be an eRAT by our definition. There is a tendency for a technology's terminology to change as it becomes mainstream (e.g. "to google" compared to "to search the world wide web"[54]). Thus more recent studies which examine the "NHS health check" may not refer to it by the computer algorithm (Qrisk) that it is based on. This review examines how eRATs change the perception of risk and uncertainty: it could be argued that when the language surrounding an eRAT fails to describe it in such terms (*i.e*. "electronic" or "computerised"), it is no longer a challenge to the way diagnoses are considered, because it is now the routine.

There was an underrepresentation of patient's views regarding eRATs: the majority of the qualitative data regarding risk and uncertainty came from clinicians in the review's studies. Research already shows If sharing diagnostic decisions with patients is to occur[55] more research is needed into the patient's perspective. There were limited findings regarding uncertainty, compared to those concerning risk. Uncertainty in medicine is an under-researched area [3,56–58] thus eRATs are influencing aspects of the consultation which are already poorly understood.

Thematic synthesis includes "translation" of the review findings into the primary studies' thematic frameworks[20]. However, perception and communication of risk and uncertainty was included in the analytical findings of only ten of the primary studies, limiting this aspect of the synthesis.

Comparison with existing literature: There are studies supporting the use of alerts regarding prescribing errors [5,59,60], and the use of similar alerts to avoid missed diagnoses is a logical step. However, this introduction of novel risk into the consultation deserves a sound evidence base which is currently missing. Mistrust of decision support systems is a recognised issue[61] and may be justified: there is limited evidence supporting eRATs in terms of improving survival and patient experience.[12,13] The role of clinical judgement in the mistrust of eRATs is a tension which has been described in other areas of evidence based medicine.[62,63] The role of gut feelings has been studied in regards to serious diagnoses [64,65] although these focused on gut feelings triggering action despite a lack of (what might be considered) objective risk, rather than decisions not to act despite risk information: the latter is likely more common in the low prevalence setting of primary care, and this area warrants further study.

The role of fear in diagnosis, although well recognised,[66] is not well researched. A more mature literature exists for fear appeals in public health messaging. Witte et al[67] has examined the Extended Parallel Process Model[68], where fear appeals produce two competing responses which interfere with each other: "danger control", where the response is to avoid the threat producing the fear, and "fear control" where the fear itself is avoided, and a health recommendation defensively resisted. Thus, fear generated by eRATs may lead to the avoidance of the eRAT, rather than any intended outcome. Fear is generally a concerning response: the systemic harms of fear on GP retention have been described[69]. Although clinical uncertainty is an inevitable part of the diagnostic process, when uncertainty becomes fear, it is likely to lead to worse overall decisions, and disengage clinicians from use of eRATs. Future research is needed into the understanding of diagnostic risk and uncertainty, from both patient and clinician's view points, how they differ and how eRATs influence the resulting communication.

Implications for future practice: This review may provide a framework through which a clinician and their patient may consider their own interactions with eRATs. The approach needs to be a nuanced

balance: to value a clinician's judgement; to be aware of the strengths and limitations of eRATs and what they bring to the consultation, whilst minimising the fear they generate for both patient and clinician.

Contributors AB: protocol, literature search, citation and full text screening, quality appraisal, data analysis, article. JF-W & BD: citation and full text screening. ES and SD: quality appraisal. ES, RC, MT and SD supervision at all stages.

Competing interests: There are no conflicts of interest declared by any author.

Funding: AB is a PhD candidate in a post funded by the University of Exeter (PhD Studentship Ref 3521) in conjunction with the ERICA trial (http://www.theericatrial.co.uk), which is funded by the Dennis and Mirelle Gillings Foundation, and receives support from the charities Cancer Research UK and Macmillan.

Data sharing statement: No additional data available

References

1. Buntinx F, Mant D, Van Den Bruel A, et al. Dealing with low-incidence serious diseases in general practice. Br J Gen Pract. 2011 Jan; 61(582):43–6.
2. Hamilton W. Cancer diagnosis in primary care. Br J Gen Pract. 2010 Feb;60(571):121–7.
3. O'Riordan M, Miguel J, Akturk Z, et al. Dealing with uncertainty in general practice: an essential skill for the general practitioner. Qual Prim Care. 2011; 19(3):175–81.
4. Maskrey N, Gordon A. Shared understanding with patients. JAMA Intern Med. 2017 Sep 1;177(9):1247–8.
5. Sutton RT, Pincock D, Baumgart DC, et al. An overview of clinical decision support systems: benefits, risks, and strategies for success. NPJ Digit Med. 2020 Feb 6;3:17.
6. Dambha-Miller H, Everitt H, Little P. Clinical scores in primary care. Br J Gen Pract. 2020 Apr 1; 70(693):163.
7. Wells PS, Anderson DR, Rodger M, et al. Excluding pulmonary embolism at the bedside without diagnostic imaging: Management of patients with suspected pulmonary embolism presenting to the emergency department by using a simple clinical model and D-dimer. Ann Intern Med. 2001 Jul 17; 135(2):98–107.
8. Royal College of Physicians London. National Early Warning Score (NEWS) 2 . Available from: https://www.rcplondon.ac.uk/projects/outputs/national-early-warning-score-news-2
9. NHS forced to admit sepsis guidance is "difficult" as GPs switch off alerts - Pulse Today. *Pulse Magazine* 2017. https://www.pulsetoday.co.uk/news/clinical-areas/infectious-diseases/nhs-forced-to-admit-sepsis-guidance-is-difficult-as-gps-switch-off-alerts/
10. Hamilton, W. Electronic Risk Assessment for Cancer for Patients in General Practice (ISRCTN22560297). *ISRCTN Registry* https://www.isrctn.com/ISRCTN22560297.
11. Shachak, A. & Reis, S. The Computer Assisted Patient Consultation. in *Clinical Technologies* 160–171 (IGI Global, 2011). doi:10.4018/978-1-60960-561-2.
12. Chima S, Reece JC, Milley K, et al. Decision support tools to improve cancer diagnostic decision making in primary care: a systematic review. Br J Gen Pract . 2019; 69(689):e809–18.
13. Riches N, Panagioti M, Alam R, et al. The effectiveness of electronic differential diagnoses (DDX) generators: A systematic review and meta-analysis. PLoS One. 2016 Mar 1;11(3):e0148991.
14. Burns A, Donnelly B, Feyi-Waboso J, Dean S. How do electronic risk assessment tools affect

the communication and understanding of diagnostic uncertainty in the primary care consultation. Do they alter how clinicians communicate risk and uncertainty? Is the understanding of uncertainty altered for either clinician or patient? . PROSPERO: International prospective register of systematic reviews. 219446. 2020. Available from: https://www.crd.york.ac.uk/prospero/display_record.php?RecordID=219446

15. Veritas Health Innovation, Melbourne A. Covidence systematic review software . Available from: www.covidence.org
16. Critical Appraisal Skills Program. CASP Qualitative Studies Checklist . 2018. Available from: https://casp-uk.b-cdn.net/
17. Garside R. Should we appraise the quality of qualitative research reports for systematic reviews, and if so, how? Innov Eur J Soc Sci Res 2013, 27:67–79.
18. International QSR. NVivo qualitative data analysis software. 2020.
19. Barnett-Page, E. & Thomas, J. Methods for the synthesis of qualitative research: A critical review. BMC Medical Research Methodology 2009, 9, 1–11.
20. Thomas, J. & Harden, A. Methods for the thematic synthesis of qualitative research in systematic reviews. BMC Medical Research Methodology 2008, 8:45
21. Paterson B, Thorne S, Canam C, et al. Meta-Study of Qualitative Health Research: A Practical Guide to Meta-Analysis and Meta-Synthesis. 1st ed. 2001.
22. Weir MC, Grimshaw JM, Mayhew A, et al. Decisions about lumping vs. splitting of the scope of systematic reviews of complex interventions are not well justified: A case study in systematic reviews of health care professional reminders. J Clin Epidemiol. 2012;65(7):756–63.
23. Babb De Villiers C., Antoniou AC, Easton D, et al. Development of cancer risk prediction models into tools for use in clinical settings. Eur J Hum Genet. 2019;26 (Supplement 1):789.
24. Chiang PP-C, Glance D, Walker J, et al. Implementing a QCancer risk tool into general practice consultations: an exploratory study using simulated consultations with Australian general practitioners. Br J Cancer. 2015;112 Suppl:S77-83.
25. Jiwa M, Skinner P, Coker, AO, et al. Implementing referral guidelines: lessons from a negative outcome cluster randomised factorial trial in general practice. BMC Fam Pr. 2006;7(PG-65):65.
26. Dikomitis L, Green T. Embedding electronic decision-support tools for suspected cancer in primary care: a qualitative study of GPs' experiences. Prim Health Care Res Dev. 2015;16(6):548–55.
27. Nieroda ME, Lophatananon A, McMillan B, et al. Online Decision Support Tool for Personalized Cancer Symptom Checking in the Community (REACT): Acceptability, Feasibility, and Usability Study. JMIR cancer. 2018;4(2):e10073.
28. Kidney E, Greenfield S, Berkman L, et al. Cancer suspicion in general practice, urgent referral, and time to diagnosis: a population-based GP survey nested within a feasibility study using information technology to flag-up patients with symptoms of colorectal cancer. BJGP Open. 2017;1(3 PG-bjgpopen17X101109)
29. Collins I, Keogh L, Steel E, et al. Development of a tailored, computerized, breast cancer risk assessment and decision support tool: What do clinicians want? J Clin Oncol. 2013;31(15 SUPPL. 1).
30. Coe AM, Ueng W, Vargas JM, et al. Usability Testing of a Web-Based Decision Aid for Breast Cancer Risk Assessment Among Multi-Ethnic Women. AMIA. Annu Symp proceedings AMIA Symp. 2017 Feb 10:411–20.
31. Keogh LA, Steel E, Weideman P, et al. Consumer and clinician perspectives on personalising breast cancer prevention information. Breast. 2019;43:39–47.
32. Kukafka R, Yi H, Xiao T, et al. Why Breast Cancer Risk by the Numbers Is Not Enough: Evaluation of a Decision Aid in Multi-Ethnic, Low-Numerate Women. J Med Internet Res . 2015;17(7):e165. 93442

For peer review only

33. Walker J, Bickerstaffe A, Hewabandu N, et al. A novel colorectal cancer risk prediction tool (CRISP): From inception to implementation. Asia Pac J Clin Oncol . 2017;13(Supplement 4):111.
34. Du X, Li S, Bhattacharyya O, et al. Identifying barriers toward implementing absolute cardiovascular risk assessment tools in primary care setting. Circulation. 2010;122(2):e363.
35. Hawkin M, Timmis A, Wilkins F, et al. Improving cardiovascular disease risk communication in NHS Health Checks: a qualitative study. BMJ Open. 2019;9(8):e026058.
36. Eaton CB, Parker DR, Craft J, et al. Using e-health to improve cholesterol management in primary care practice. J Med Pract Manage. 2009;24(4):224–30.
37. Peiris DP, Joshi R, Webster RJ, et al. An electronic clinical decision support tool to assist primary care providers in cardiovascular disease risk management: development and mixed methods evaluation. J Med Internet Res. 2009;11(4):e51.
38. Short D, Frischer M. Barriers to the adoption of computerised decision support systems in general practice consultations: A qualitative study of GPs' perspectives. Int J Med Inform. 2004;73(4):357–62.
39. Short D, Frischer M, Bashford J. The development and evaluation of a computerised decision support system for primary care based upon "patient profile decision analysis". Inform Prim Care. 2003;11(4):195–202.
40. Porat T, Marshall IJ, Sadler E, et al. Collaborative design of a decision aid for stroke survivors with multimorbidity: A qualitative study in the UK engaging key stakeholders. BMJ Open. 2019;9(8):e030385.
41. Porat T, Liao Z, Curcin V, et al. Engaging Stakeholders in the Design and Usability Evaluation of a Decision Aid to Improve Secondary Stroke Prevention. Stud Health Technol Inform. 2018;247:765–9.
42. Colombet I, Dart T, Leneveut L, et al. A computer decision aid for medical prevention: a pilot qualitative study of the Personalized Estimate of Risks (EsPeR) system. BMC Med Inform Decis Mak. 2003;3:13.
43. Harry ML, Truitt AR, Saman DM, et al. Barriers and facilitators to implementing cancer prevention clinical decision support in primary care: a qualitative study. BMC Health Serv Res. 2019;19(1):534.
44. Saver B, Mazor K, Hargraves J, et al. Inaccurate risk perceptions and individualized risk estimates by patients with type 2 diabetes. J Am Board Fam Med. 2014;27(4 PG-510–9):510–9.
45. Crawford F, Bekker HL, Young M, et al. General practitioners' and nurses' experiences of using computerised decision support in screening for diabetic foot disease: implementing Scottish Clinical Information - Diabetes Care in routine clinical practice. Inform Prim Care. 2010;18(4):259–68.
46. Vail L, Adams A, Gilbert E, et al. Investigating mental health risk assessment in primary care and the potential role of a structured decision support tool, GRiST. Ment Health Fam Med. 2012;9(1):57–67. 08183CFB6CC4BDEDC0D64F5
47. Richardson S, Feldstein D, McGinn T, et al. Live Usability Testing of Two Complex Clinical Decision Support Tools: Observational Study. JMIR Hum Factors. 2019 Apr 15;6(2):e12471.
48. Voruganti T., O'Brien M, Straus S, et al. Primary care physicians' perspectives on computer-based health risk assessment tools for chronic diseases: a mixed methods study. J Innov Heal informatics . 2015;22(3):333–9.
49. Porat T, Delaney B, Kostopoulou, O. The impact of a diagnostic decision support system on the consultation: perceptions of GPs and patients. BMC Med Inform Decis Mak . 2017;17(1):79.
50. McParland C, Cooper M. Differential Diagnosis Decision Support Systems in Primary and Out-of-Hours Care: A Qualitative Analysis of the Needs of Key Stakeholders in Scotland. J Prim Care Community Health. 2019;10:2150132719829315.

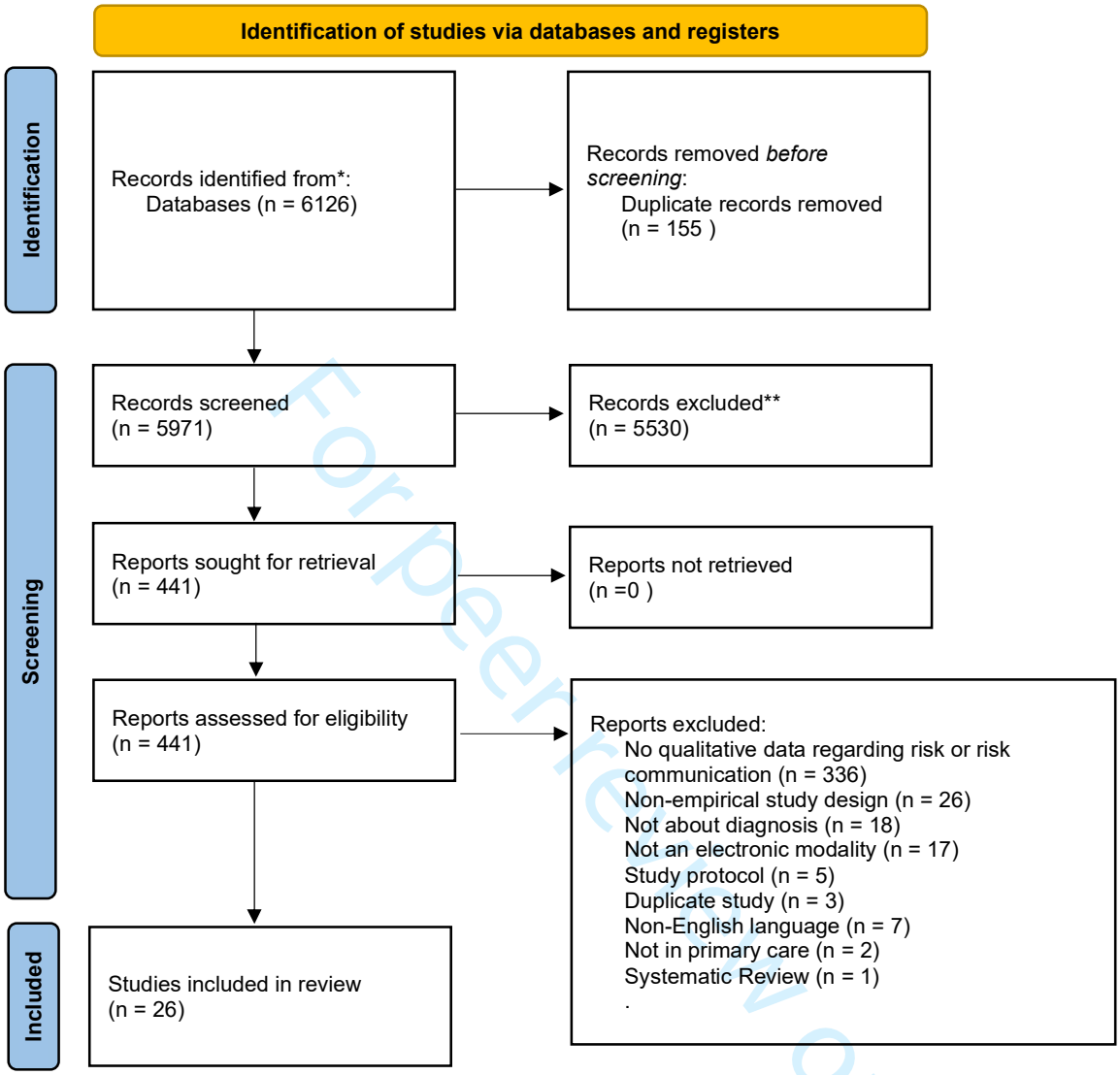
For peer review only

51. Eccles DW, Arsal G. The think aloud method: what is it and how do I use it? Qual Res Sport Exerc Heal. 2017 Aug 8;9(4):514–31.
52. Singh H, Schiff GD, Graber ML, et al. The global burden of diagnostic errors in primary care. BMJ Qual Saf. 2016/08/16. 2017 Jun;26(6):484–94.
53. Elwyn G, Frosch D, Thomson R, et al. Shared decision making: a model for clinical practice. J Gen Intern Med. 2012 Oct 27(10):1361–7.
54. Glance D. How technology is changing language and the way we think about the world. The Conversation. 2015 Jan 5
55. Berger ZD, Brito JP, Ospina NS, et al. Patient centred diagnosis: sharing diagnostic decisions with patients in clinical practice. BMJ, 2017 Nov 1; 359:j4218.
56. Spence JD. The need for clinical judgement in the application of evidence-based medicine. BMJ Evidence-Based Med . 2020 Oct 1;25(5):172 LP – 177.
57. Chartash D, Sassoon D, Muthu N. Physicians in the Era of Automation: The Case for Clinical Expertise. MDM policy Pract . 2019 Jul 1;4(2):2381468319868968.
58. Friedemann Smith C, Drew S, Ziebland S, et al. Understanding the role of GPs' gut feelings in diagnosing cancer in primary care: a systematic review and meta-analysis of existing evidence. Br J Gen Pract. 2020 Sep 1;70(698):E612–21.
59. Hjertholm P, Moth G, Ingeman ML, et al. Predictive values of GPs' suspicion of serious disease: a population-based follow-up study. Br J Gen Pract. 2014; 64(623):e346 LP-e353.
60. Heath I. Role of fear in overdiagnosis and overtreatment—an essay by Iona Heath. BMJ. 2014 Oct 24;349:g6123.
61. Jones C, Thornton J, Wyatt JC. Enhancing trust in clinical decision support systems: a framework for developers. BMJ Health and Care Informatics. 2021 Jun 1;28(1):e100247.
62. Spence, J. D. The need for clinical judgement in the application of evidence-based medicine. BMJ Evidence-Based Medicine 2020, 25, 172–177
63. Chartash, D., Sassoon, D. & Muthu, N. Physicians in the Era of Automation: The Case for Clinical Expertise. MDM policy & practice 2019, 4, 2381468319868968
64. Friedemann Smith, C., Drew, S., Ziebland, S. & Nicholson, B. D. Understanding the role of GPs' gut feelings in diagnosing cancer in primary care: a systematic review and meta-analysis of existing evidence. Br J Gen Pract. 2020; 70, E612–E621
65. Hjertholm P, Moth G, Ingeman ML, et al. Predictive values of GPs' suspicion of serious disease: a population-based follow-up study. Br J Gen Pract. 2014 Jun 1;64(623):e346 LP-e353.
66. Heath, I. Role of fear in overdiagnosis and overtreatment—an essay by Iona Heath. BMJ, 2014; 349, g6123
67. Witte K, Allen M. A Meta-Analysis of Fear Appeals: Implications for Effective Public Health Campaigns. Heal Educ Behav. 2000 Oct 1;27(5):591–615.
68. Leventhal H. Findings and Theory in the Study of Fear Communications. In: Advances in Experimental Social Psychology, editor Berkowitz L. Academic Press; 1970. p. 119–86.
69. Campbell, J. L. *et al.* Policies and strategies to retain and support the return of experienced GPs in direct patient care: the ReGROUP mixed-methods study. Health Services and Delivery Research 2019, vol. 7, 14

For peer review only

For peer review only

Fig 1: Flow Diagram for systematic searches of databases and registers



For peer review only

Table 1: Characteristics of Included Studies

Study Author and Date	Country of Subjects	Diagnosis of interest	Research Subjects	eRAT exposure	Data Collection Methodology	Data Analysis	Risk/uncertainty theme in 1° analysis	CASP Quality Assessment
Babb de Villiers 2019	UK and international	Cancer	Clinicians	Workshop	Workshop	Thematic Analysis	No	**2**
Chiang 2015	Australia	Cancer	Clinicians	Simulated Clinic	Semi-structured Interviews	NPT	No	**8**
Coe 2016	USA	Breast Cancer	Patients	Workshop/Usability testing	"Morae" software recording.	Framework Analysis (implied not stated)	No	**8**
Collins 2013	Australia	Breast cancer	Clinicians	Workshop	Focus groups	Unclear	No	**4**
Colombet 2003	France	Cardiovascular Disease, Cancer, MH risks	Clinicians	Workshop/Usability testing	Focus group	Framework Analysis (implied not stated)	Yes	**6**
Crawford 2010	UK	DM foot disease	Clinicians	Real World Use	Semi-structured interviews	Thematic Analysis	No	**8**
Dikimitis 2015	UK	Cancer	Clinicians	Real World Use	Semi-structured interviews	NPT	No	**8**
Du 2010	China	Cardiovascular Disease	Clinicians	Workshop/Usability testing	Semi Structured focus groups	Inductive coding	No	**3**
Eaton 2009	USA	Cardiovascular Disease	Patients	Real World Use	Focus Groups	Unclear	Yes	**4**
Harry 2019	USA	Cancer Cardiovascular Disease	Clinicians	Real World Use	Semi-structured interviews	Grounded Theory	No	**9**

For peer review only

Hawking 2019	UK	Cardiovascular Disease	Patients	Real World Use	Semi-structured interviews	Thematic Analysis	Yes	**8**
Jiwa 2006	USA	Cancer	Clinicians	Real World Use	Semi structured interviews	Thematic Analysis	No	**7**
Keogh 2019	Australia and New Zealand	Breast Cancer	Clinicians and Patients	Workshop/Usability Testing	Focus Group	Thematic Analysis (implied not stated)	No	**8**
Kidney 2017	UK	Colorectal Cancer	Clinicians and Practice Managers	Real World Use	Interviews	Thematic Analysis	Yes	**6**
Kukafka 2015	USA	Breast Cancer	Patients	Real World Use	Focus groups	Thematic Analysis (implied not stated)	Yes	**8**
McParland 2019	UK	DDx	Clinicians	Real World Use	Focus groups	Thematic Analysis	No	**8**
Nieroda 2018	UK	Cancer	Clinicians and Patients	Workshop/Usability Testing	Focus groups, think aloud use of website, open ended questionnaires	Thematic Analysis	No	**8**
Peiris 2009	Australia	CVD	Clinicians	Real World Use	Semi-structured interviews	Thematic Analysis	Yes	**8**
Porat 2017	UK	DDx	Clinicians	Simulated Clinic	Semi-structured interviews	Thematic Analysis	No	**7**
Porat 2018	UK	Stroke	Clinicians	Workshop/Usability Testing	Semi-structured interviews	Thematic Analysis	Yes	**3**

For peer review only

Porat 2019	UK	Stroke	Clinicians, Patients, and other Stakeholders	Workshop/Usability testing	Focus groups, Interviews and usability evaluations.	Thematic Analysis	Yes	**5**
Richardson 2019	USA	Pharyngitis and Cough	Clinicians	Real World Use	Interview, video and audio of consultation	Constant Comparative Method	No	**8**
Saver 2014	USA	Diabetes complications	Patients	Real World Use	Interviews	general inductive approach	Yes	**5**
Short 2003	UK	Stroke	Clinicians	Simulated Clinic	Qualitative interview and questionnaire	Unclear	No	**7**
Short 2004	UK	Stroke	Clinicians	Workshop/Usability testing	Interviews	Constant Comparative Method	Yes	**8**
Vail 2012	UK	Mental Health risk	Clinicians	Workshop/Usability testing	Interviews, focus groups	Thematic Analysis	No	**8**
Voruganti 2015	Canada	Multiple	Clinicians	Workshop/Usability testing	Focus Groups and usability testing	Grounded theory	No	**8**
Walker 2017	Australia	Colorectal Cancer	Clinicians and Practice Managers	Simulated Clinic	Interviews, videoed simulated consultations analysed	Framework Analysis	No	**7**

For peer review only

Table 2: Assessment of quality of studies with the CASP tool

Study ID	1. Was there a clear statement of the aims of the research?	2. Is a qualitative methodology appropriate?	3. Was the research design appropriate to address the aims of the research?	4. Was the recruitment strategy appropriate to the aims of the research?	5. Was the data collected in a way that addressed the research issue?	6. Has the relationship between researcher and participants been adequately considered?	7. Have ethical issues been taken into consideration?	8. Was the data analysis sufficiently rigorous?	9. Is there a clear statement of findings?	10. How valuable is the research?
Babb de Villiers 2019	Yes	Yes	No	No	Can't tell	No	No	No	No	No
Chiang 2015	Yes	Yes	Yes	Yes	Yes	Can't tell	Yes	Yes	Yes	Yes
Coe 2016	Yes	Yes	Can't tell	Yes	Yes	No	Yes	Yes	Yes	Yes
Collins 2013	Yes	Yes	Yes	Can't tell	Can't tell	Can't tell	Can't tell	Can't tell	Yes	Yes
Colombet 2003	Yes	Yes	Yes	Yes	Yes	Can't tell	Yes	Yes	Yes	Yes
Crawford 2010	Yes	Yes	Yes	Yes	Yes	Can't tell	Yes	Yes	Yes	Yes
Dikimitis 2015	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Du 2010	Yes	Yes	Yes	Can't tell	Can't tell	Can't tell	Can't tell	Can't tell	Can't tell	Can't tell
Eaton 2009	Yes	Yes	Can't tell	Can't tell	Yes	No	No	Can't tell	Yes	Yes
Harry 2019	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Hawking 2019	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Jiwa 2006	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Can't tell	Can't tell
Keogh 2019	Yes	Yes	Yes	Yes	Yes	Can't tell	Yes	Yes	Yes	Yes
Kidney 2017	Yes	Yes	Yes	Can't tell	Yes	No	Yes	Can't tell	Yes	Yes
Kukafka 2015	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
McParland 2019	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Nieroda 2018	Yes	Yes	Yes	Yes	Yes	Can't tell	Yes	Yes	Yes	Yes

For peer review only

Peiris 2009	Yes	Yes	Yes	Yes	Yes	Can't tell	Yes	Yes	Yes	Yes
Porat 2017	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Porat 2018	No	Yes	Can't tell	Yes	Yes	No	No	No	Yes	Can't tell
Porat 2019	Yes	Yes	Yes	Yes	Yes	No	Can't tell	Can't tell	Yes	Can't tell
Richardson 2019	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Saver 2014	Yes	Yes	Can't tell	Yes	Can't tell	No	Yes	Yes	Yes	Can't tell
Short 2003	Yes	Yes	Yes	Can't tell	Yes	Can't tell	Yes	Yes	Yes	Yes
Short 2004	Yes	Yes	Can't tell	Yes	Yes	No	No	No	Yes	Yes
Vail 2012	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Voruganti 2015	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Walker 2017	Yes	Yes	Yes	Yes	Can't tell	Can't tell	Yes	Yes	Yes	Yes

For peer review only

Table 3: Illustrative Quotes to reinforce Analytical Themes

Theme		Quote (all clinician comments, unless stated)	Study
Novel Risk	*Alerts to new risk*	*"It's just electronic highlight, makes you think even if you immediately dismiss it, at least that millisecond you've thought about it, and I think that is going to be useful at some point, but for how many people I don't know"*	Dikomitis 2015
		"Helpful to refresh your memory about things that are rare…we mainly stick to the phrase: 'common things are common"	Porat 2017
	Alerts to alternative risks	*"Yes, I must admit ovarian didn't come so high up, I had already fixated on the bowel stuff, which maybe I shouldn't have. This really said hey, consider ovarian as well. So again that was useful, yes"*	Chiang 2015
		"I think it widens your diagnosis, like the guy that had COPD and then ended up with AS. I would get to the diagnosis, but it would be longer without the tool."	Porat 2017
	Alerts to risk factor	*"the patient can see very clearly, [and say] 'Wow, smoking contributes a lot to cardiovascular risk, I do actually want to think about smoking cessation'"*	Voruganti 2015
		"I don't think it's (the eRAT) altered practice. I think it's kept some of the not so hard red flag symptoms in people's minds…"	Kidney 2017
Risk Refinement	*Orientation of risk (within guidelines/clinical pathways)*	*"Normally I'd get a few investigations, get the results back and then based on that say do we need to do something, or I refer this on based on that. But I guess if I have a calculator saying it's higher risk, it might prompt me to make a referral to a specialist a bit earlier."*	Chiang 2015
		"Think the biggest use I got, because you presented with very constitutional symptoms that a lot of things could present with, this was useful for me to think what's the good next step and what should I think of. So I was thinking in my mind more"	Chiang 2015
	More accurate risk, reduction of uncertainty	*For clinicians, the main perceived benefit was iPrevent's provision of accurate and reliable risk estimates for both ten year and lifetime risk, and standardized risk management advice for women*	Keogh 2019, Researcher comment
		"It is helpful in that it reminds me that the risk of stroke goes up substantially as you get older so your benefit (of aspirin) increases"	Short 2003
	Simplification of risk, reduction of uncertainty	*"Well I suppose you could think about the common comorbidities, so hypertension and stroke, AF (atrial fibrillation) and stroke, diabetes and stroke and you could think about not necessarily an algorithm but a sort of stepwise*	Porat 2019

For peer review only

		prioritisation about what you should think about in terms of the patient's total management, you know, which would be the most important area of focus?"	
	Introduction of new risk knowledge	*"Other colleagues have said as well we're far more aware of thrombocytosis and increased platelets. We weren't aware that tended to increase the score for increased cancer risk"*	Dikomitis 2015
		"The learning aspect is very important, since this system is based on evidence and evidence can change"	Porat 2019
Communication	*Patient to Clinician Communication*	*"I think it's important to be informed and to get there as early as possible. To go and see the GP, and if there's a tool perhaps, you know, to help me do this, that may be of interest"*	Nieroda 2018 Patient Comment
	Risk focuses conversations	*"it usually stops a lot of the meandering dialogue that you'd otherwise engage in…. [initiate] that discussion in terms of what their risk factors are"*	Voruganti 2015
		"So having that piece of paper there gave you that conversation: 'Well now we've finished everything, let's look at this.'"	Peiris 2009
	Risk justifies action	*"But your heart is beating kinda fast, you've had a fever last night...the recommendation would be to get a chest x-ray today."*	Richardson 2019
		"OK, so our little risk calculator here is recommending that we would swab you for strep throat, and I agree with that."	Richardson 2019
	Risk justifies inaction	*"…there's a whole lot of people and this is your chance out of them of cancer…"*	Walker 2007
		"I think FRAX [is] useful because [it] can cut down on unnecessary investigations"	Voruganti 2015
	Improves understanding of risk	*"I think the biggest impact is that it changed the way I talked about what I was doing with patients, in that it made it a much more slick, neat package to describe the normal screening that you do for risk management. And so I felt it was easier to deliver some description of where they're at now. And from their point of view, I mean it's hard to know, but they seemed to understand that it was a multifactorial thing, rather than just being one of those single disease problems…. The thing that I don't really know, that I guess would be useful, is what they think when they walk out the door, what they actually understand of what I've said."*	Pieris 2009
		"With our Framingham tool you get…, a yellow bar. The average person's 10-year risk is…two little yellow bars, and your [the patient's] average risk is…maybe five little yellow bars'"	Voruganti 2015

For peer review only

		"I like this one [referring to the color bar].... I mean, everyone knows that red means danger, so if they're heading towards this one, it's a lot more visual, the impact...."	Peiris 2009
Autonomy	*Empowered by knowing personalised risk*	*"Anything that can be personalised or tailored, so you don't feel it's this off the shelf thing that you're being given, you know… you sit with your doctor and it's not just a case of giving out a leaflet, but actually let's have a look at your personal data"*	Porat 2019
		"I think the benefit is you're empowering yourself with information, you're possibly changing what you can change"	Keogh 2019
		"Anything that gets people to have a think about their health is a good thing, anything at all."	Walker 2007
Fear	*Direct fear of diagnosis*	*"Well I just kind of—a heart attack is a scariest thing to me. And I think it just kind of got me wondering and thinking and got me too nervous about the whole thing"*	Eaton 2009, Patient comment
		"And I thought, 'Wow! Patients coming with these tummy pains,' even if I'm thinking, 'She's got a 10% chance,' to actually think, 'Wow! Listen, you got a 38% chance of cancer.' General public are very—I would find that confronting and the general public have a shocking record at understanding risk."	Chiang 2015
	Disturbs normal practice	*"It's completely different to how I normally work, unsettling and confusing, I'm not quite sure where I am, to what extent I'm following a template and how I take history."*	Porat 2017
		"I personally don't think tools help me too much, interfere with my process of thought, would have done better without it, it was unnatural"	Porat 2017
	Removes focus and control of consultation	*"it might bring up a lot more other issues that they [patients] weren't originally aware of and the discussion might actually… be less directed"*	Voruganti 2015
		'Yeah. If I start saying to them, 'Oh god, your risk of having gastro-oesophageal cancer is one in three'y that seems way over the top, back pedalling from that, the consultation would've gone on for another 20 minutes or 30 minutes since I tried to explain to him, no. He doesn't have it'	Chiang 2015
	Lack of agency over risk	*The most commonly expressed concern was the potential for the risk assessment tool to miss something that was unknown and outside their control*	Kukafka 2015, Researcher Comment
	Medico-legal fear	*"Quite a few partners were worried about any medico-legal implications with*	Dikomitis 2015

For peer review only

		that. I think the one criticism was having the list of patient with PPVs. No one really liked that. That was worrying. We're worried about if patients knew that you had a list of them with the risk and you hadn't acted on it, what would be the implications? That was probably a point that put people off, really"	
		Some also felt that overriding a DDDSS would make them vulnerable to litigation	McParland 2019, Researcher Comment
Mistrust	*eRAT over-triggers*	*"Inevitably it will come up with a diagnosis of cancer. When the patient walks through the door . . . quite clearly, quickly, we can see this is not a serious diagnosis"*	McParland 2019
		"It gives information which, as it's blandly presented, you go, "How did you get that?…" I got a couple of people where I got a 20% number and you go, "Oh that's madness, that's not you," and often because it's based on single digit information…like a single blood pressure."	Peiris 2009
	eRAT misses things	*A consistent theme in our analysis was uncertainty among participants that all factors were accounted for in determining their breast cancer risk*	Kukafka 2015, Researcher comment
	eRAT not applicable to their patient	*In half of the patient visits, patients reported some piece of information, typically as a part of the history of present illness that raised a question for the coders of whether the tool was applicable to their clinical condition.*	Richardson 2019, Researcher comment
		they felt obliged to use the tool for the purpose of the study and their suspicion of cancer was unnaturally heightened	Chiang 2015
	Preference for diagnostic tests over risk scores	*"I would like to know if I have the gene mutation. So for me to say that being my mom had it [breast cancer] and take a pill and prevent it and not knowing if I have the gene doesn't make sense to me."*	Kukafka 2015
		"Colonoscopy was commonly referred to as the 'gold standard' for CRC screening. GPs were more likely to recommend colonoscopy if the patient wanted one or they had previously been recommended one by a gastroenterologist, even if the patient was at average risk."	Walker 2007, Researcher comment
	Preference for clinical judgement	*"… more experienced GPs were less likely to agree with the underlying concept and continued to maintain strong belief in their clinical intuition."*	Chiang 2015, Researcher comment
		"Patients don't follow flow charts, in my experience"	McParland 2019

For peer review only

		…that's just the tricky bit of saying [to the patient] that this is what's [sic] the recommendation but we don't always follow what recommendations are…	Walker 2007
	Patient preference for self knowledge	*"I think the model is incorrect because I know I am going to last longer than 10 years, because I do take care of myself"*	Saver 2014

For peer review only

ENTREQ checklist (Enhancing transparency in reporting the synthesis of qualitative research) *

No. Item	Guide questions/description	Reported on Page #
1. Aim	State the research question the synthesis addresses	Title & 4
2. Synthesis methodology	Identify the synthesis methodology or theoretical framework which underpins the synthesis, and describe the rationale for choice of methodology (e.g. meta-ethnography, thematic synthesis, critical interpretive synthesis, grounded theory synthesis, realist synthesis, meta-aggregation, meta-study, framework synthesis)	5
3. Approach to searching	Indicate whether the search was pre-planned (comprehensive search strategies to seek all available studies) or iterative (to seek all available concepts until they theoretical saturation is achieved)	5
4. Inclusion criteria	Specify the inclusion/exclusion criteria (e.g. in terms of population, language, year limits, type of publication, study type)	5
5. Data sources	Describe the information sources used (e.g. electronic databases (MEDLINE, EMBASE, CINAHL, psycINFO), grey literature databases (digital thesis, policy reports), relevant organisational websites, experts, information specialists, generic web searches (Google Scholar) hand searching, reference lists) and when the searches conducted; provide the rationale for using the data sources	5
6. Electronic Search strategy	Describe the literature search (e.g. provide electronic search strategies with population terms, clinical or health topic terms, experiential or social phenomena related terms, filters for qualitative research, and search limits)	Appendix
7. Study screening methods	Describe the process of study screening and sifting (e.g. title, abstract and full text review, number of independent reviewers who screened studies)	5
8. Study characteristics	Present the characteristics of the included studies (e.g. year of publication, country, population, number of participants, data collection, methodology, analysis, research questions)	Table 1
9. Study selection results	Identify the number of studies screened and provide reasons for study exclusion (e,g, for comprehensive searching, provide numbers of studies screened and reasons for exclusion indicated in a figure/flowchart; for iterative searching describe reasons for study exclusion and inclusion based on modifications to the research question and/or contribution to theory development)	5-6 and Figure 1
10. Rationale for appraisal	Describe the rationale and approach used to appraise the included studies or selected findings (e.g. assessment of conduct (validity and robustness), assessment of reporting (transparency), assessment of content and utility of the findings)	4

For peer review only

11. Appraisal items	State the tools, frameworks and criteria used to appraise the studies or selected findings (e.g. Existing tools: CASP, QARI, COREQ, Mays and Pope [25]; reviewer developed tools; describe the domains assessed: research team, study design, data analysis and interpretations, reporting)	5
12. Appraisal process	Indicate whether the appraisal was conducted independently by more than one reviewer and if consensus was required	5
13. Appraisal results	Present results of the quality assessment and indicate which articles, if any, were weighted/excluded based on the assessment and give the rationale	Table 2
14. Data extraction	Indicate which sections of the primary studies were analysed and how were the data extracted from the primary studies? (e.g. all text under the headings "results /conclusions" were extracted electronically and entered into a computer software)	5
15. Software	State the computer software used, if any	5
16. Number of reviewers	Identify who was involved in coding and analysis	5
17. Coding	Describe the process for coding of data (e.g. line by line coding to search for concepts)	5
18. Study comparison	Describe how were comparisons made within and across studies (e.g. subsequent studies were coded into pre-existing concepts, and new concepts were created when deemed necessary)	5
19. Derivation of themes	Explain whether the process of deriving the themes or constructs was inductive or deductive	5
20. Quotations	Provide quotations from the primary studies to illustrate themes/constructs, and identify whether the quotations were participant quotations of the author's interpretation	Table 3
21. Synthesis output	Present rich, compelling and useful results that go beyond a summary of the primary studies (e.g. new interpretation, models of evidence, conceptual models, analytical framework, development of a new theory or construct)	6-8

* Reference: Tong A, Flemming K, McInnes E, Oliver SA, Craig J. Enhancing transparency in reporting the synthesis of qualitative research: ENTREQ. BMC Medical Research Methodology 2012, 12:181.

Search Strategy

#	Database	Search term	Results
1	Medline	("electronic risk assessment*").ti,ab	13
2	Medline	(electronic ADJ3 ("risk assessment*" OR RAT OR eRAT OR RATs OR eRATs OR diagnosis)).ti,ab	459

For peer review only

3	Medline	(comput* ADJ3 ("risk assessment*" OR RAT OR eRAT OR RATs OR eRATs OR diagnosis)).ti,ab	8879
4	Medline	"DECISION SUPPORT TECHNIQUES"/	19374
5	Medline	"DECISION SUPPORT SYSTEMS, CLINICAL"/ OR "DIAGNOSIS, COMPUTER-ASSISTED"/	29078
6	Medline	"RISK ASSESSMENT"/	249398
7	Medline	(electronic OR comput*).ti,ab	805823
8	Medline	(6 AND 7)	10675
9	Medline	("decision support").ti,ab	13295
10	Medline	("prediction tool*").ti,ab	3009
11	Medline	(qcancer).ti,ab	10
12	Medline	("risk prediction").ti,ab	8010
13	Medline	("computer guided consultation").ti,ab	2
24	Medline	(1 OR 2 OR 3 OR 4 OR 5 OR 8 OR 9 OR 10 OR 11 OR 12 OR 13)	82373
14	Medline	exp "GENERAL PRACTICE"/	73872
15	Medline	"PRIMARY HEALTH CARE"/	73929
16	Medline	"GENERAL PRACTITIONERS"/	7252
17	Medline	"PHYSICIANS, FAMILY"/	16158
18	Medline	("general practi*").ti,ab	38963
19	Medline	("primary health care").ti,ab	19769
20	Medline	("primary care").ti,ab	106822
21	Medline	(GP OR GPs).ti,ab	55107

For peer review only

22	Medline	("family physician*").ti,ab	14422
23	Medline	("family practice").ti,ab	7465
25	Medline	(14 OR 15 OR 16 OR 17 OR 18 OR 19 OR 20 OR 21 OR 22 OR 23)	274818
26	Medline	(24 AND 25)	3066
27	EMBASE	("electronic risk assessment*").ti,ab	30
28	EMBASE	(electronic ADJ4 ("risk assessment*" OR RAT OR eRAT OR RATs OR eRATs OR diagnosis)).ti,ab	811
29	EMBASE	(comput* ADJ4 ("risk assessment*" OR RAT OR eRAT OR RATs OR eRATs OR diagnosis)).ti,ab	12523
30	EMBASE	exp "DECISION SUPPORT SYSTEM"/ OR "COMPUTER ASSISTED DIAGNOSIS"/	61278
31	EMBASE	"RISK ASSESSMENT"/	534341
32	EMBASE	(electronic OR comput*).ti,ab	1181113
33	EMBASE	(31 AND 32)	24951
34	EMBASE	("decision support").ti,ab	17689
35	EMBASE	("prediction tool*").ti,ab	4544
36	EMBASE	(qcancer).ti,ab	14
37	EMBASE	("risk prediction").ti,ab	12919
38	EMBASE	("computer guided consultation").ti,ab	8
39	EMBASE	(27 OR 28 OR 29 OR 30 OR 33 OR 34 OR 35 OR 36 OR 37 OR 38)	117361
40	EMBASE	"GENERAL PRACTICE"/	76245
41	EMBASE	exp "PRIMARY HEALTH CARE"/	162467

For peer review only

42	EMBASE	"GENERAL PRACTITIONER"/	94915
43	EMBASE	("general practi*").ti,ab	96563
44	EMBASE	("primary health care").ti,ab	23923
45	EMBASE	("primary care").ti,ab	146773
46	EMBASE	(GP OR GPs).ti,ab	79393
47	EMBASE	("family physician*").ti,ab	17848
48	EMBASE	("family practice").ti,ab	7988
49	EMBASE	(40 OR 41 OR 42 OR 43 OR 44 OR 45 OR 46 OR 47 OR 48)	415195
50	EMBASE	(39 AND 49)	4595
51	CINAHL	("electronic risk assessment*").ti,ab	7
52	CINAHL	(electronic ADJ4 ("risk assessment*" OR RAT OR eRAT OR RATs OR eRATs OR diagnosis)).ti,ab	279
53	CINAHL	(comput* ADJ4 ("risk assessment*" OR RAT OR eRAT OR RATs OR eRATs OR diagnosis)).ti,ab	2066
54	CINAHL	"DECISION SUPPORT SYSTEMS, CLINICAL"/	4554
55	CINAHL	exp "DECISION SUPPORT TECHNIQUES"/	9645
56	CINAHL	"DIAGNOSIS, COMPUTER ASSISTED"/	4031
57	CINAHL	"RISK ASSESSMENT"/	87369
58	CINAHL	(electronic OR comput*).ti,ab	184706
59	CINAHL	(57 AND 58)	4195
60	CINAHL	("decision support").ti,ab	5671
61	CINAHL	("prediction tool*").ti,ab	678

For peer review only

62	CINAHL	(qcancer).ti,ab	8
63	CINAHL	("risk prediction").ti,ab	2809
64	CINAHL	("computer guided consultation").ti,ab	1
65	CINAHL	(51 OR 52 OR 53 OR 54 OR 55 OR 56 OR 59 OR 60 OR 61 OR 62 OR 63 OR 64)	29549
66	CINAHL	"FAMILY PRACTICE"/	22717
67	CINAHL	"PHYSICIANS, FAMILY"/	17517
68	CINAHL	"PRIMARY HEALTH CARE"/	56524
69	CINAHL	("general practi*").ti,ab	26608
70	CINAHL	("primary health care").ti,ab	10032
71	CINAHL	(GP OR GPs).ti,ab	18016
72	CINAHL	("family physician*").ti,ab	5248
73	CINAHL	("family practice").ti,ab	1902
74	CINAHL	(66 OR 67 OR 68 OR 69 OR 70 OR 71 OR 72 OR 73)	111915
75	CINAHL	(65 AND 74)	1406

Web of Science Search

(“electronic risk assessment” OR electronic NEAR/3 ("risk assessment*" OR RAT OR eRAT OR RATs OR eRATs OR diagnosis) OR comput* NEAR/3 ("risk assessment*" OR RAT OR eRAT OR RATs OR eRATs OR diagnosis) OR “decision support” OR "prediction tool*" OR qcancer OR “risk prediction” OR "computer guided consultation") AND (“General practi*” OR “Primary Health Care” OR “Family Physician*” OR “Primary Care” OR GP OR GPs OR “Family Practice”)

For peer review only

BMJ Open

How do electronic risk assessment tools affect the communication and understanding of diagnostic uncertainty in the primary care consultation? A systematic review and thematic synthesis

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-060101.R1
Article Type:	Original research
Date Submitted by the Author:	11-May-2022
Complete List of Authors:	Burns, Alex; University of Exeter, College of Medicine and Health; PenARC, Donnelly, Brian; University of Exeter Medical School Feyi-Waboso, Joshua; University of Exeter Medical School Shephard, Elizabeth; University of Exeter, College of Medicine and Health Calitri, Raff; University of Exeter, College of Medicine and Health Tarrant, Mark; University of Exeter, College of Medicine and Health Dean, Sarah; PenARC; University of Exeter, College of Medicine and Health
Primary Subject Heading:	General practice / Family practice
Secondary Subject Heading:	Diagnostics, Qualitative research
Keywords:	PRIMARY CARE, QUALITATIVE RESEARCH, Risk management < HEALTH SERVICES ADMINISTRATION & MANAGEMENT

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

How do electronic risk assessment tools affect the communication and understanding of diagnostic uncertainty in the primary care consultation? A systematic review and thematic synthesis

Dr Alex Burns MBBS, MGHP, BA(cantab). GP Partner, Threespires Medical Practice, Truro. PhD Student, University of Exeter, College of Medicine and Health. ORCID 0000-0002-9931-9198

Brian Donnelly. Medical Student, University of Exeter Medical School

Joshua Feyi-Waboso. Medical Student, University of Exeter Medical School. ORCID 0000-0002-5417-4298

Dr Elizabeth Shephard PhD, MSc, BSc. Research Fellow, University of Exeter, College of Medicine and Health. ORCID 0000-0002-3610-3680

Dr Raff Calitri PhD, MSc, BSc. Research Fellow, University of Exeter, College of Medicine and Health. ORCID 0000-0003-0889-4670

Dr Mark Tarrant PhD, BA(hons). Associate Professor, University of Exeter, College of Medicine and Health. ORCID 0000-0002-7194-1428

Professor Sarah Dean, PhD, MSc, BSc. Professor, University of Exeter, College of Medicine and Health. ORCID 0000-0002-3682-5149

corresponding author: Dr Alex Burns, ab1265@exeter.ac.uk

For peer review only

How do electronic risk assessment tools affect the communication and understanding of diagnostic uncertainty in the primary care consultation? A systematic review and thematic synthesis

Alex Burns, Brian Donnelly, Joshua Feyi-Waboso, Elizabeth Shephard, Raff Calitri, Mark Tarrant and Sarah Dean

Abstract

Objectives: To conduct a systematic review and synthesise qualitative research of electronic risk assessment tools (eRATs) in primary care, examining how they affect the communication and understanding of diagnostic risk and uncertainty. eRATs are computer-based algorithms designed to help clinicians avoid missing important diagnoses, pick up possible symptoms early and facilitate shared decision making.

Design: Systematic search, using predefined criteria of the published literature and synthesis of the qualitative data, using Thematic Synthesis. Database searches on 27th November 2019 were of MEDLINE, EMBASE, CINAHL and Web of Science, and a secondary search of the references of included articles. Included studies were those involving electronic risk assessment or decision support, pertaining to diagnosis in primary care, where qualitative data were presented. Non-empirical studies and non-English language studies were excluded. 5971 unique studies were identified of which 441 underwent full text review. 26 studies were included for data extraction. A further 2 were found from citation searches. Quality appraisal was via the CASP tool. Data extraction was via line by line coding. A Thematic Synthesis was performed.

Setting: Primary Care

Results: Electronic risk assessment tools included differential diagnosis suggestion tools, tools which produce a future risk of disease development or recurrence, or calculate a risk of current undiagnosed disease. Analytical themes were developed to describe separate aspects of the clinical consultation where risk and uncertainty are both central and altered via the use of an eRAT: “Novel risk”, “Risk refinement”, “Autonomy”,” Communication”, “Fear” and “Mistrust”.

Conclusion: Despite little evidence on the effectiveness of electronic risk assessment tools in primary care, their role is likely to increase. Understanding how they influence consultations may help both clinicians and patients in their interaction with eRATs.

Registered review with Prospero, CRD219446

Strengths and Limitations:

- This is the first review to address the impact of eRATs in the primary care consultation
- There is inconsistency in nomenclature surrounding these tools
- There is a lack of data from patients in the primary studies
- There are likely biases in the original studies.

For peer review only

Introduction

Diagnosis in primary care is complex and challenging. The main input, a patient's history and examination, can be variable, imperfect and subjective. Disease prevalence is lower than other health-care settings[1,2]. GPs are able to process and communicate risk and accept uncertainty in diagnosis:[3] Doing this in a way which allows a shared understanding of medicine with a patient is one of the seminal challenges of primary care.[4]

Clinical Decision Support Systems are designed to "enhance medical decisions with targeted clinical knowledge, patient information, and other health information"[5]. "Electronic Risk Assessment Tools" (hereafter referred to as eRATs) are decision support systems which help with diagnosis. They are software where information about a patient is processed electronically to produce an output pertinent to a diagnosis, either immediately or in the future. The output can be a quantitative risk of a specific diagnosis, or a more qualitative output: prompting the clinician to consider a particular diagnosis or list of possible diagnoses. eRATs can enable extraction of data previously inputted into primary care IT systems, and can prompt users to input missing data. They can incorporate "pop ups" during a consultation to avoid missing important diagnoses.[6,7] They have a range of stated aims: making diagnosis more evidence based and objective; avoiding missed diagnoses; or facilitating shared decision making.

eRATs tend to target diagnoses which are both serious and benefit from early intervention,[6,7] an obvious example being cancer. To achieve adequate sensitivity these algorithms tend towards a low positive predictive value.[6,7] As eRATs use more complex calculations, invisible to the clinician, and insert themselves into the clinician consultation more proactively, they may come to represent a "third actor" in the consultation, alongside the patient and clinician.[8] They have the potential to change the consultation dynamic and diagnostic thinking process. Although there have been systematic reviews conducted on the effectiveness of eRATs[9,10], there have been no systematic reviews using qualitative methods examining risk and uncertainty in their use.

Diagnosis in primary care tends to be the outcome of clinical reasoning, which can be considered using the theories of Situativity:[11] that diagnosis will depend on other actors (as well as the clinician) present during the consultation, and their interaction with multiple environmental inputs; in our case eRATs. Within this clinical situation we are interested in the communication of risk, which is the best proxy we have of the understanding of risk. The Theory of Communicative Rationality [12] may help our understanding - communication can be emancipative: an attempt to achieve mutual understanding (as in shared decision making); or strategic, where the intention of the communication is to achieve a particular aim of an individual, such as calculative manipulation of others.

Aim

This study aims to synthesise qualitative research on eRATs in the primary care consultation, examining how they affect the communication and understanding of the risk of a particular disease, and uncertainty when used making diagnoses.

Methods

For peer review only

Study Selection

Searches of MEDLINE, EMBASE, CINAHL and Web of Science were performed. The search strategy[13] targeted studies involving a primary care population, where risk assessment or decision support tools are used for, where qualitative data are reported regarding risk or uncertainty. Non-empirical studies and non-English language studies were excluded, as well as studies outside of primary care, or where qualitative data were not presented. Using Covidence software[14], three authors (AB, JF-W and BD) independently screened titles and abstracts, and then full text for relevance. Disputes were resolved on discussion between authors. The initial 5971 unique studies that were identified reduced to 441 on title/abstract review. These underwent full text review, which yielded 26 individual studies, and one systematic review[9] (Figure 1). These were hand searched for further citations, which yielded 2 additional studies. Key eRATs embedded into System1, a mainstream UK IT system, were identified and the developers emailed (where qualitative studies had not already been identified) to see if there were unpublished data available. This did not identify further studies. Thus 28 independent studies were included for data extraction (supplementary table1).

Quality appraisal Data Extraction

The 28 studies were assessed for quality (AB, ES and SD) using the CASP critical appraisal checklist[15]. The CASP checklist was designed to be used as educational pedagogic tool and not as a scoring tool for systematic reviews. A point was assigned if the study met one of the 10 statements making up the CASP checklist. Discrepancies were resolved on discussion between the authors (supplementary 2). None were excluded from analysis due to low quality[16]. Data extraction was performed by AB. Quotes and researcher statements regarding risk and uncertainty were extracted electronically and coded into NVivo software.[17]

Thematic Synthesis

Thematic Synthesis was chosen as it is based in Scientific Realism[18]: the purpose of this review is to examine how clinicians and patients understanding of risk and uncertainty change with the use of eRATs. For the results to be broadly applicable, the underlying assumption is that the understanding of risk, and this communication is consistent, or consistent within a range.

Descriptive codes were developed by AB. Codes were described in the initial studies coded, and subsequent studies coded into these concepts, with new descriptive codes being created, refined or ordered hierarchically when deemed necessary. These descriptive codes were discussed with other authors (SD, ES, RC and MT). Thematic Synthesis involves the “translation of concepts from one study to another”[19]. The ten studies with analytical findings pertaining to risk or uncertainty were examined. The developed descriptive codes were applied to each to explore new coding approaches. The descriptive codes were then developed inductively into analytical themes describing separate aspects of the clinical consultation where risk and uncertainty is both central to, and altered via, the use of an eRAT. Selective quotes are used to illustrate the richness of data and to provide support for these themes.

Patient and public involvement

Patients and the public were not involved in the review

Research ethics approval

This study did not receive or require ethics approval, as it does not involve human or animal participants

Results

Three conceptually different forms of eRATs were identified: those which produced a risk of an immediate, or "clinical" diagnosis, (e.g., the risk of a current bacterial tonsillitis); those which produced a risk of a future or "prognostic" diagnosis, (e.g., the risk of cardiovascular disease in the next decade); and those which produced a list of potential alternative diagnoses, or "differential diagnosis generators". This is an important distinction because it gets to the heart of what diagnosis is: separating the affected from the non-affected. This means defining illness, and the concept of current illness is distinct to the threat of future illness.[20]

Except for a single study,[21] those concerning eRATs for clinical diagnosis collected data from clinicians rather than patients. Studies of prognostic eRATs were more orientated towards the communication of risk to patients. The studies examining differential diagnosis generators had data from both clinicians and patients.

The data was synthesised data the three conceptual forms, and themes explored to see if there were findings consistent between them[22]. This acknowledges the aspects of the different eRATs which are the same and contributory to the aims of the review.

Included studies had publication dates ranging from 2003 to 2019. [21,23–49] There was a broad range of targeted diagnoses, as well as data collection and analytical methods. There was also a range of nomenclature used to describe the eRAT being used. Details are available in supplementary table 1. Only ten of the studies reported themes or findings pertaining to risk or uncertainty in their analysis.[27,33–35,37–40,42,49]

Analytical Themes:

Six analytical themes are presented below with supportive quotes (further supportive quotes are presented in supplementary table 3. Quotes are from clinicians, unless specified otherwise.

"Novel risk". A new risk is introduced into the consultation, either as an unconsidered diagnosis or an unexpected high risk.[24,26,28,33,46,47] It brings the risk to the forefront of the consultation and ensures that it is adequately considered.

"It's just electronic highlight, makes you think even if you immediately dismiss it, at least that millisecond you've thought about it, and I think that is going to be useful at some point, but for how many people I don't know."[26]

"I think it's important to be informed and to get there as early as possible. To go and see the GP, and if there's a tool perhaps, you know, to help me do this, that may be of interest." Patient Comment[21]

The eRAT can raise the risk of alternative diagnoses:

For peer review only

"Yes, I must admit ovarian didn't come so high up, I had already fixated on the bowel stuff, which maybe I shouldn't have. This really said hey, consider ovarian as well. So again that was useful, yes."[24]

This theme arose mainly from studies regarding clinical diagnosis, and differential diagnosis generators, and arose mainly from data regarding clinicians views.

"Risk refinement". In this theme, the risk is processed into a more accurate assessment. It was generated from studies concerning eRATs for prognostic and clinical diagnosis, and mainly from clinicians.

It enables clinicians: "t*o take complex, competing information and bring it together".*[36]

This refinement of risk serves several purposes. It can give an objective single point estimate of risk.[28,36] it can take complex risk factors and combine them into a single risk output,[37] and it can orientate the clinician towards guidelines and clinical pathways, triggering them to add missing information to the eRAT,[39] suggesting the next diagnostic tests which are appropriate[24] to reduce uncertainty and improve diagnostic accuracy.

"Communication". eRATs can assist clinicians to communicate risk to patients, and in patient-facing eRATs, to help patients communicate with their clinicians[21]. They can provide a focus point for a consultation.[39,46] Clinicians can use the eRAT output to justify their next action, be this referral or further testing particularly in eRATs for clinical diagnosis.[24,45] Where the risk output is low they can justify inaction.[24,31]

"OK, so our little risk calculator here is recommending that we would swab you for strep throat, and I agree with that."[45]

"I think FRAX [is] useful because [it] can cut down on unnecessary investigations."[46]

Prognostic eRATs, can emphasise the level of risk, with a preference for colour displays or other qualitative methods of showing risk.[39,46] eRAT outputs can be used as a motivator to take treatments or change behaviour.[38]

"…you're possibly changing what you can change" Patient comment[28]

"Autonomy". The eRAT output is a personalised risk pertaining to a patient's health, enabling them to make informed decisions. This is regarded as of innate benefit.[28,37]

"Anything that gets people to have a think about their health is a good thing, anything at all."[31]

The fact that the eRAT has an output which not generic, but personalised to the patient's data, was perceived to be an important component of this benefit amongst clinicians and patients across the range of eRATs examined in this review.

"Fear". The eRAT could create anxiety or disquiet in both patient and clinician. This can occur at several levels. The first is the fear of a serious diagnosis from the eRAT. Patients have understandable concern about their own health and welfare:[34]

"If I knew I was from a high risk category, and I'm pretty sure I'm not, I reckon I would find that so stressful. I would just worry every time there was a pimple", Patient comment[28]

For peer review only

clinicians share this fear, with an overlay of concern about poor understanding of risk among patients:

"And I thought, 'Wow! Patients coming with these tummy pains,' even if I'm thinking, 'She's got a 10% chance,' to actually think, 'Wow! Listen, you got a 38% chance of cancer.' General public are very—I would find that confronting and the general public have a shocking record at understanding risk."[24]

A second fear from clinicians regarding differential diagnosis generators, and clinical diagnosis eRATs, was of the tool interrupting how they normally think about diagnosis.[47,48] The eRAT could distort how they handled diagnostic risk and unwanted risk suggestions made clinicians uncertain. There was a fear that the risk outputs could lead to a loss of control of the consultation, as the uncertainties raised by the eRATs override other agendas.[24,46]

Finally, there were fears expressed regarding the medico-legal consequences of eRATs. Clinicians expressed concern that any eRAT output, if not acted on, could act as evidence against them[26,48] and make them vulnerable to litigation. Here though, there was a negative case in the data, with one clinician reporting the eRAT was reassuring: that it acted as a "backstop", reducing fear of missing a diagnosis.

"So although it can cause irritation, the flip side of that is you can relax a little bit more and that you don't have to remember absolutely everything (…) You know, how to sort the wheat from the chaff. So although people might think it's irritating, it can be reassuring."[26]

"Mistrust". There was a theme of mistrust in the risk output of the eRAT. Some clinicians felt that the eRAT algorithm was flawed: because the trigger was set at an incorrect level of risk;[25,48] that it relied on a single parameter, and changes to that parameter could disproportionately change the output;[39] that the inputs themselves were out of date or flawed in some way; or that the outputs were outside the control of patient and clinician. There was then a suspicion that the eRAT could be either under or over reporting a risk.

"Inevitably it will come up with a diagnosis of cancer. When the patient walks through the door... quite clearly, quickly, we can see this is not a serious diagnosis"[48]

Although clinicians could use the eRAT output to justify not performing further investigations, this tended to be in high risk situations where further testing was unnecessary before diagnosis and action. In situations where there was a low risk output from the eRAT, both clinicians and patients preferred diagnostic tests over the output of an eRAT.[31,49] Here, a diagnostic test would be ordered despite the eRAT suggesting a risk low enough that testing was unnecessary. The test was trusted and the eRAT discounted.

There was faith in personal judgement over the eRAT. Patients preferentially trusting their knowledge of their own bodies,[28,42] clinicians trusting their clinical acumen or gut instinct.[24,31,40,48] Over-ruling an eRAT with clinical acumen was seen as good medicine:

"Patients don't follow flow charts, in my experience."[48]

Discussion

For peer review only

Summary: This review found that eRATs can influence diagnosis, and the communication of its risks and uncertainties in differing ways. Situativity[11] includes the idea of Distributed Cognition[50], where reasoning is spread between several actors or tools. An eRAT introducing "Novel" risk in a consultation could be seen as an example of this, and could have the potential to overcome the cognitive errors of misdiagnosis (such as premature closure).[51] Shared Decision Making[52] could be seen as a combination of the themes of "Communication", "Autonomy" and "Risk Refinement", helping realise the Emancipative Communication as described by Habermas[12]. In addition, the "Risk Refinement" function of an eRAT has the potential to increase the consistency of diagnostic decision making, which might improve adherence to guidelines, and play a part in cost containment.[5] These more positive themes could be undermined by those of "Mistrust" and "Fear", where the risk outputs of the eRAT are disregarded, or seen as something that could challenge or disturb the communication of diagnostic risk.

Strengths and limitations: This review is the first to address how eRATs affect understanding and communication of risk and uncertainty in the primary care consultation. The broad scope of this review means a range of eRATs have been synthesised. Some themes presented derive a greater part of their data from eRATs of a particular conceptual design ("clinical", "prognostic" or "differential diagnosis", as described in the results section), which could challenge their transferability across all eRATs. This review was intended to look at the wider clinician and patient interaction (in terms of understanding and communication) with eRATs and the findings are broadly transferable, which supports the analytical approach taken.

Another challenge concerns the nomenclature surrounding these tools: we used search terms to capture all studies where information was processed electronically to produce an output pertinent to a diagnosis, either immediately or in the future: here we have called them eRATs. The language used by the originators is often different and has changed over time. eRATs can also be referred to by other names, such as Diagnostic Decision Support Systems, Electronic Screening Tools and Risk Prediction Tools. The review may have missed interventions which, if examined, might be an eRAT by our definition.

There was limited data from patient's views regarding eRATs: the majority of the qualitative data regarding risk and uncertainty came from clinicians in the review's studies, and data from patients was less conceptually rich. If sharing diagnostic decisions with patients is to occur[53] more research is needed into the patient's perspective. There were limited findings regarding uncertainty, compared to those concerning risk. Uncertainty in medicine is an under-researched area [3,54–56] thus eRATs are influencing aspects of the consultation which are already poorly understood.

Thematic synthesis includes "translation" of the review findings into the primary studies' thematic frameworks. This involves "*the process of taking concepts from one study and recognising the same concepts in another study*". [19] However, understanding and communication of risk and uncertainty was included in the analytical findings of only ten of the primary studies, limiting this aspect of the synthesis. Single author data extraction is also a limitation, as the findings may be more prone to author bias.

Comparison with existing literature: There are studies supporting the use of alerts regarding prescribing errors [5,57,58], and the use of similar alerts to avoid missed diagnoses is a logical step. However, this introduction of novel risk into the consultation deserves a sound evidence base which

For peer review only

is currently missing. Mistrust of decision support systems is a recognised issue[59] and may be justified: there is limited evidence supporting eRATs in terms of improving survival and patient experience.[12,13] The role of clinical judgement in the mistrust of eRATs is a tension which has been described in other areas of evidence based medicine.[60,61] Part of clinical judgement is a clinicians' gut feelings. The role of gut feelings has been studied in regards to serious diagnoses [62,63] although these focused on gut feelings triggering action despite a lack of (what might be considered) objective risk, rather than decisions not to act despite risk information: the latter is likely more common in the low prevalence setting of primary care, and this area warrants further study.

The role of fear in diagnosis, although well recognised,[64] is not well researched. A more mature literature exists for fear appeals in public health messaging. Witte et al[65] has examined the Extended Parallel Process Model[66], where fear appeals produce two competing responses which interfere with each other: "danger control", where the response is to avoid the threat producing the fear, and "fear control" where the fear itself is avoided, and a health recommendation defensively resisted. Thus, fear generated by eRATs may lead to the avoidance of the eRAT, rather than any intended outcome. Fear is generally a concerning response: the systemic harms of fear on GP retention have been described[67]. Although clinical uncertainty is an inevitable part of the diagnostic process, when uncertainty becomes fear, it is likely to lead to worse overall decisions, and disengage clinicians from use of eRATs. Future research is needed into the understanding of diagnostic risk and uncertainty, from both patient and clinician's view points, how they differ and how eRATs influence the resulting communication.

Implications for future research and practice: Future research is needed into the understanding of diagnostic risk and uncertainty, from both patient and clinician's view points, how they differ and how eRATs influence the resulting communication. More evidence is needed examining how eRATs influence a clinicians clinical reasoning, particularly when the eRAT challenges clinical judgement. This review may provide a framework through which a clinician and their patient may consider their own interactions with eRATs. The approach needs to be a nuanced balance: to value a clinician's judgement; to be aware of the strengths and limitations of eRATs and what they bring to the consultation, whilst minimising the fear they generate for both patient and clinician.

Contributors AB: protocol, literature search, citation and full text screening, quality appraisal, data analysis, article. JF-W & BD: citation and full text screening. ES and SD: quality appraisal. ES, RC, MT and SD supervision at all stages.

Competing interests: There are no conflicts of interest declared by any author.

Funding: AB is a PhD candidate in a post funded by the University of Exeter (PhD Studentship Ref 3521) in conjunction with the ERICA trial (http://www.theericatrial.co.uk), which is funded by the Dennis and Mirelle Gillings Foundation, and receives support from the charities Cancer Research UK and Macmillan.

Data sharing statement: No additional data available

Figure 1 caption: Flow Diagram for systematic searches of databases and registers

References

1. Buntinx F, Mant D, Van Den Bruel A, et al. Dealing with low-incidence serious diseases in general practice. Br J Gen Pract. 2011 Jan; 61(582):43–6.
2. Hamilton W. Cancer diagnosis in primary care. Br J Gen Pract. 2010 Feb;60(571):121–7.
3. O'Riordan M, Miguel J, Akturk Z, et al. Dealing with uncertainty in general practice: an essential skill for the general practitioner. Qual Prim Care. 2011; 19(3):175–81.
4. Maskrey N, Gordon A. Shared understanding with patients. JAMA Intern Med. 2017 Sep 1;177(9):1247–8.
5. Sutton RT, Pincock D, Baumgart DC, et al. An overview of clinical decision support systems: benefits, risks, and strategies for success. NPJ Digit Med. 2020 Feb 6;3:17.
6. NHS forced to admit sepsis guidance is "difficult" as GPs switch off alerts - Pulse Today. *Pulse Magazine* 2017. https://www.pulsetoday.co.uk/news/clinical-areas/infectious-diseases/nhs-forced-to-admit-sepsis-guidance-is-difficult-as-gps-switch-off-alerts/
7. Hamilton, W. Electronic Risk Assessment for Cancer for Patients in General Practice (ISRCTN22560297). *ISRCTN Registry* https://www.isrctn.com/ISRCTN22560297.
8. Shachak, A. & Reis, S. The Computer Assisted Patient Consultation. in *Clinical Technologies* 160–171 (IGI Global, 2011). doi:10.4018/978-1-60960-561-2.
9. Chima S, Reece JC, Milley K, et al. Decision support tools to improve cancer diagnostic decision making in primary care: a systematic review. Br J Gen Pract . 2019; 69(689):e809–18.
10. Riches N, Panagioti M, Alam R, et al. The effectiveness of electronic differential diagnoses (DDX) generators: A systematic review and meta-analysis. PLoS One. 2016 Mar 1;11(3):e0148991.
11. Durning, S. J., Artino, A. R. J., Schuwirth, L. & van der Vleuten, C. Clarifying Assumptions to Enhance Our Understanding and Assessment of Clinical Reasoning. Academic Medicine. 2013, 88.
12. Habermas, J. The Theory of Communicative Action,1984, Beacon Press.
13. Burns A, Donnelly B, Feyi-Waboso J, Dean S. How do electronic risk assessment tools affect the communication and understanding of diagnostic uncertainty in the primary care consultation. Do they alter how clinicians communicate risk and uncertainty? Is the understanding of uncertainty altered for either clinician or patient? . PROSPERO: International prospective register of systematic reviews. 219446. 2020. Available from: https://www.crd.york.ac.uk/prospero/display_record.php?RecordID=219446
14. Veritas Health Innovation, Melbourne A. Covidence systematic review software . Available from: www.covidence.org
15. Critical Appraisal Skills Program. CASP Qualitative Studies Checklist . 2018. Available from: https://casp-uk.b-cdn.net/
16. Garside R. Should we appraise the quality of qualitative research reports for systematic reviews, and if so, how? Innov Eur J Soc Sci Res 2013, 27:67–79.
17. International QSR. NVivo qualitative data analysis software. 2020.
18. Barnett-Page, E. & Thomas, J. Methods for the synthesis of qualitative research: A critical review. BMC Medical Research Methodology 2009, 9, 1–11.
19. Thomas, J. & Harden, A. Methods for the thematic synthesis of qualitative research in systematic reviews. BMC Medical Research Methodology 2008, 8:45
20. Coggon, D., Barker, D. & Rose, G. Epidemiology for the Uninitiated, 2003. Wiley.
21. Nieroda ME, Lophatananon A, McMillan B, et al. Online Decision Support Tool for Personalized Cancer Symptom Checking in the Community (REACT): Acceptability, Feasibility, and Usability Study. JMIR cancer. 2018;4(2):e10073.
22. Weir MC, Grimshaw JM, Mayhew A, et al. Decisions about lumping vs. splitting of the scope of systematic reviews of complex interventions are not well justified: A case study in systematic reviews of health care professional reminders. J Clin Epidemiol. 2012;65(7):756–

For peer review only

63.
23. Babb De Villiers C., Antoniou AC, Easton D, et al. Development of cancer risk prediction models into tools for use in clinical settings. Eur J Hum Genet. 2019;26 (Supplement 1):789.
24. Chiang PP-C, Glance D, Walker J, et al. Implementing a QCancer risk tool into general practice consultations: an exploratory study using simulated consultations with Australian general practitioners. Br J Cancer. 2015;112 Suppl:S77-83.
25. Jiwa M, Skinner P, Coker, AO, et al. Implementing referral guidelines: lessons from a negative outcome cluster randomised factorial trial in general practice. BMC Fam Pr. 2006;7(PG-65):65.
26. Dikomitis L, Green T. Embedding electronic decision-support tools for suspected cancer in primary care: a qualitative study of GPs' experiences. Prim Health Care Res Dev. 2015;16(6):548–55.
27. Kidney E, Greenfield S, Berkman L, et al. Cancer suspicion in general practice, urgent referral, and time to diagnosis: a population-based GP survey nested within a feasibility study using information technology to flag-up patients with symptoms of colorectal cancer. BJGP Open. 2017;1(3 PG-bjgpopen17X101109)
28. Collins I, Keogh L, Steel E, et al. Development of a tailored, computerized, breast cancer risk assessment and decision support tool: What do clinicians want? J Clin Oncol. 2013;31(15 SUPPL. 1).
29. Coe AM, Ueng W, Vargas JM, et al. Usability Testing of a Web-Based Decision Aid for Breast Cancer Risk Assessment Among Multi-Ethnic Women. AMIA. Annu Symp proceedings AMIA Symp. 2017 Feb 10:411–20.
30. Keogh LA, Steel E, Weideman P, et al. Consumer and clinician perspectives on personalising breast cancer prevention information. Breast. 2019;43:39–47.
31. Kukafka R, Yi H, Xiao T, et al. Why Breast Cancer Risk by the Numbers Is Not Enough: Evaluation of a Decision Aid in Multi-Ethnic, Low-Numerate Women. J Med Internet Res . 2015;17(7):e165. 93442
32. Walker J, Bickerstaffe A, Hewabandu N, et al. A novel colorectal cancer risk prediction tool (CRISP): From inception to implementation. Asia Pac J Clin Oncol . 2017;13(Supplement 4):111.
33. Du X, Li S, Bhattacharyya O, et al. Identifying barriers toward implementing absolute cardiovascular risk assessment tools in primary care setting. Circulation. 2010;122(2):e363.
34. Hawkin M, Timmis A, Wilkins F, et al. Improving cardiovascular disease risk communication in NHS Health Checks: a qualitative study. BMJ Open. 2019;9(8):e026058.
35. Eaton CB, Parker DR, Craft J, et al. Using e-health to improve cholesterol management in primary care practice. J Med Pract Manage. 2009;24(4):224–30.
36. Peiris DP, Joshi R, Webster RJ, et al. An electronic clinical decision support tool to assist primary care providers in cardiovascular disease risk management: development and mixed methods evaluation. J Med Internet Res. 2009;11(4):e51.
37. Short D, Frischer M. Barriers to the adoption of computerised decision support systems in general practice consultations: A qualitative study of GPs' perspectives. Int J Med Inform. 2004;73(4):357–62.
38. Short D, Frischer M, Bashford J. The development and evaluation of a computerised decision support system for primary care based upon "patient profile decision analysis". Inform Prim Care. 2003;11(4):195–202.
39. Porat T, Marshall IJ, Sadler E, et al. Collaborative design of a decision aid for stroke survivors with multimorbidity: A qualitative study in the UK engaging key stakeholders. BMJ Open. 2019;9(8):e030385.
40. Porat T, Liao Z, Curcin V, et al. Engaging Stakeholders in the Design and Usability Evaluation of a Decision Aid to Improve Secondary Stroke Prevention. Stud Health Technol Inform. 2018;247:765–9.

For peer review only

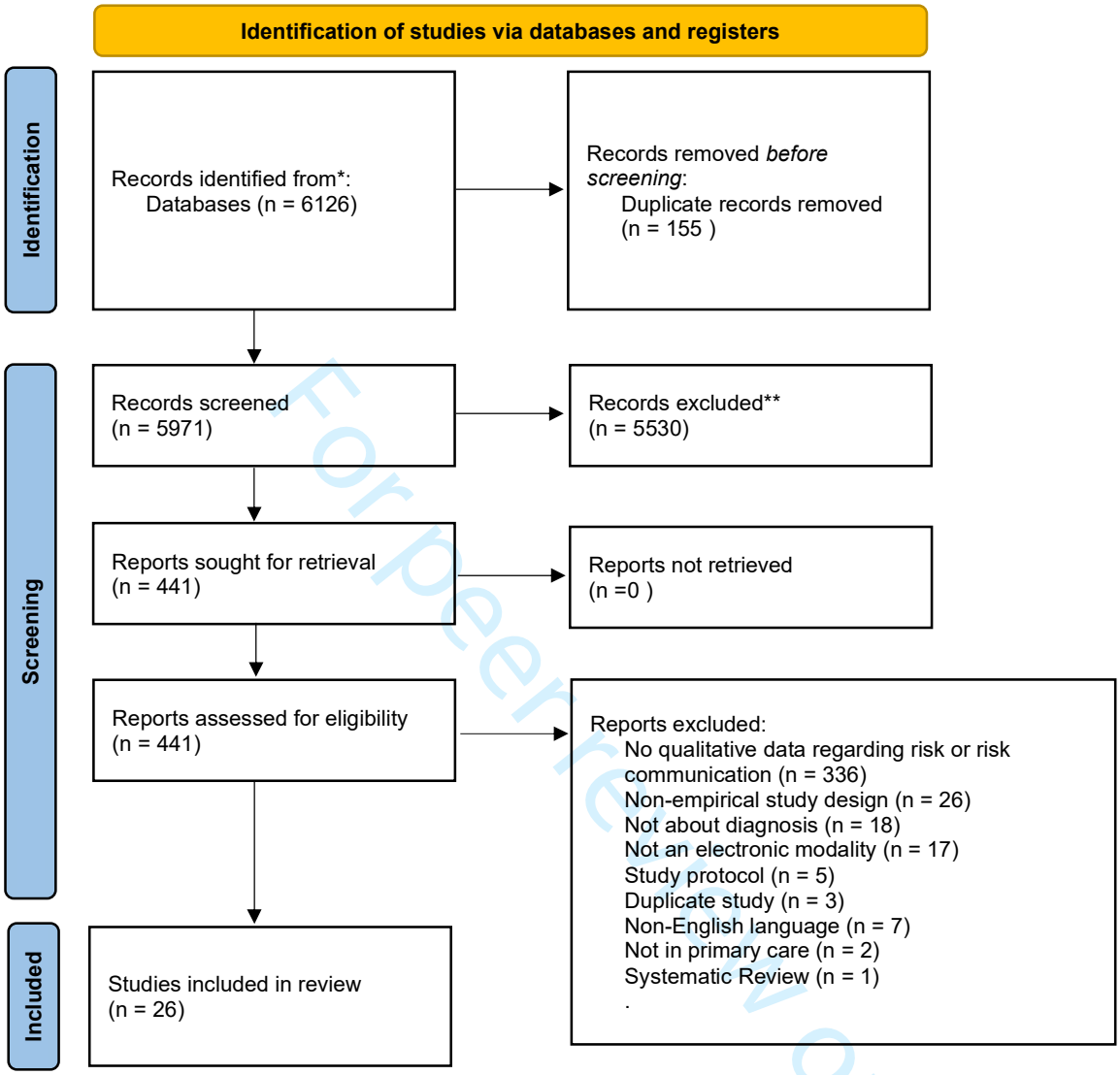
41. Colombet I, Dart T, Leneveut L, et al. A computer decision aid for medical prevention: a pilot qualitative study of the Personalized Estimate of Risks (EsPeR) system. BMC Med Inform Decis Mak. 2003;3:13.
42. Harry ML, Truitt AR, Saman DM, et al. Barriers and facilitators to implementing cancer prevention clinical decision support in primary care: a qualitative study. BMC Health Serv Res. 2019;19(1):534.
43. Saver B, Mazor K, Hargraves J, et al. Inaccurate risk perceptions and individualized risk estimates by patients with type 2 diabetes. J Am Board Fam Med. 2014;27(4 PG-510–9):510–9.
44. Crawford F, Bekker HL, Young M, et al. General practitioners' and nurses' experiences of using computerised decision support in screening for diabetic foot disease: implementing Scottish Clinical Information - Diabetes Care in routine clinical practice. Inform Prim Care. 2010;18(4):259–68.
45. Vail L, Adams A, Gilbert E, et al. Investigating mental health risk assessment in primary care and the potential role of a structured decision support tool, GRiST. Ment Health Fam Med. 2012;9(1):57–67. 08183CFB6CC4BDEDC0D64F5
46. Richardson S, Feldstein D, McGinn T, et al. Live Usability Testing of Two Complex Clinical Decision Support Tools: Observational Study. JMIR Hum Factors. 2019 Apr 15;6(2):e12471.
47. Voruganti T., O'Brien M, Straus S, et al. Primary care physicians' perspectives on computer-based health risk assessment tools for chronic diseases: a mixed methods study. J Innov Heal informatics . 2015;22(3):333–9.
48. Porat T, Delaney B, Kostopoulou, O. The impact of a diagnostic decision support system on the consultation: perceptions of GPs and patients. BMC Med Inform Decis Mak. 2017;17(1):79.
49. McParland C, Cooper M. Differential Diagnosis Decision Support Systems in Primary and Out-of-Hours Care: A Qualitative Analysis of the Needs of Key Stakeholders in Scotland. J Prim Care Community Health. 2019;10:2150132719829315.
50. Merkebu, J. *et al.* Situativity: a family of social cognitive theories for understanding clinical reasoning and diagnostic error. Diagnosis 7, 2020, 169–176.
51. Singh H, Schiff GD, Graber ML, et al. The global burden of diagnostic errors in primary care. BMJ Qual Saf. 2016/08/16. 2017 Jun;26(6):484–94.
52. Elwyn G, Frosch D, Thomson R, et al. Shared decision making: a model for clinical practice. J Gen Intern Med. 2012 Oct 27(10):1361–7.
53. Berger ZD, Brito JP, Ospina NS, et al. Patient centred diagnosis: sharing diagnostic decisions with patients in clinical practice. BMJ, 2017 Nov 1; 359:j4218.
54. Hatch, S. Uncertainty in medicine. BMJ, 2017, 357:j2180
55. Hillen, MA, Gutheil, CM., Strout, TD, et al. Tolerance of uncertainty: Conceptual analysis, integrative model, and implications for healthcare. Social Science and Medicine, 2017. vol. 180 62–75
56. Cox, CL, Miller, BM, Kuhn, I, et al. Diagnostic uncertainty in primary care: what is known about its communication, and what are the associated ethical issues? Family Practice, 2021. doi:10.1093/fampra/cmab023.
57. Coma, E, Medina, M, Mendez, L et al. Effectiveness of electronic point-of-care reminders versus monthly feedback to improve adherence to 10 clinical recommendations in primary care: a cluster randomized clinical trial. BMC Med Inform Decis Mak*, 2019* **19**, 245.
58. Schedlbauer, A, Prasad, V, Mulvaney, C. et al. What Evidence Supports the Use of Computerized Alerts and Prompts to Improve Clinicians' Prescribing Behavior? J of the Am Med Inform Ass, 2009, 16, 531–538.
59. Jones C, Thornton J, Wyatt JC. Enhancing trust in clinical decision support systems: a framework for developers. BMJ Health and Care Informatics. 2021 Jun 1;28(1):e100247.

For peer review only

60. Spence, J. D. The need for clinical judgement in the application of evidence-based medicine. BMJ EBM 2020, 25, 172–177
61. Chartash, D., Sassoon, D. & Muthu, N. Physicians in the Era of Automation: The Case for Clinical Expertise. MDM policy & practice 2019, 4,
62. Friedemann Smith, C., Drew, S., Ziebland, S. & Nicholson, B. D. Understanding the role of GPs' gut feelings in diagnosing cancer in primary care: a systematic review and meta-analysis of existing evidence. Br J Gen Pract. 2020; 70, E612–E621
63. Hjertholm P, Moth G, Ingeman ML, et al. Predictive values of GPs' suspicion of serious disease: a population-based follow-up study. Br J Gen Pract. 2014 Jun 1;64(623):e346 LP-e353.
64. Heath, I. Role of fear in overdiagnosis and overtreatment—an essay by Iona Heath. BMJ, 2014; 349, g6123
65. Witte K, Allen M. A Meta-Analysis of Fear Appeals: Implications for Effective Public Health Campaigns. Heal Educ Behav. 2000 Oct 1;27(5):591–615.
66. Leventhal H. Findings and Theory in the Study of Fear Communications. In: Advances in Experimental Social Psychology, editor Berkowitz L. Academic Press; 1970. p. 119–86.
67. Campbell, J. L. *et al.* Policies and strategies to retain and support the return of experienced GPs in direct patient care: the ReGROUP mixed-methods study. Health Services and Delivery Research 2019, vol. 7, 14

For peer review only

Fig 1: Flow Diagram for systematic searches of databases and registers



For peer review only

Supplementary Table 1: Characteristics of Included Studies

Study Author and Date	Country of Subjects	Diagnosis of interest	eRAT form	Nomenclature used in study	Research Participant	eRAT exposure	Data Collection Methodology	Data Analysis	Risk/uncertainty theme in 1° analysis	CASP Quality Assessment
Babb de Villiers 2019	UK and international	Cancer	Prognostic	Risk prediction models	Clinicians	Workshop	Workshop	Thematic Analysis	No	**2**
Chiang 2015	Australia	Cancer	Clinical	Risk tool	Clinicians	Simulated Clinic	Semi-structured Interviews	NPT	No	**8**
Coe 2016	USA	Breast Cancer	Prognostic	Risk Assessment	Patients	Workshop/Usability testing	"Morae" software recording.	Framework Analysis (implied not stated)	No	**8**
Collins 2013	Australia	Breast cancer	Prognostic	Risk assessment and decision support	Clinicians	Workshop	Focus groups	Unclear	No	**4**
Colombet 2003	France	Cardiovascular Disease, Cancer, MH risks	Prognostic	Personalised Estimate of Risk	Clinicians	Workshop/Usability testing	Focus group	Framework Analysis (implied not stated)	Yes	**6**
Crawford 2010	UK	DM foot disease	Prognostic	Risk Assessment Tool	Clinicians	Real World Use	Semi-structured interviews	Thematic Analysis	No	**8**
Dikimitis 2015	UK	Cancer	Clinical	Electronic Risk Assessment Tools	Clinicians	Real World Use	Semi-structured interviews	NPT	No	**8**
Du 2010	China	Cardiovascular Disease	Prognostic	Risk Assessment Tools	Clinicians	Workshop/Usability testing	Semi Structured focus groups	Inductive coding	No	**3**

For peer review only

Eaton 2009	USA	Cardiovascular Disease	Prognostic	Interactive Risk Communication and Decision Support	Patients	Real World Use	Focus Groups	Unclear	Yes	**4**
Harry 2019	USA	Cancer Cardiovascular Disease	Prognostic	Clinical Decision Support	Clinicians	Real World Use	Semi-structured interviews	Grounded Theory	No	**9**
Hawking 2019	UK	Cardiovascular Disease	Prognostic	Risk Report	Patients	Real World Use	Semi-structured interviews	Thematic Analysis	Yes	**8**
Jiwa 2006	USA	Cancer	Clinical	Interactive referral proforma	Clinicians	Real World Use	Semi structured interviews	Thematic Analysis	No	**7**
Keogh 2019	Australia and New Zealand	Breast Cancer	Prognostic	Risk assessment and management support tool	Clinicians and Patients	Workshop/Usability Testing	Focus Group	Thematic Analysis (implied not stated)	No	**8**
Kidney 2017	UK	Colorectal Cancer	Clinical	Information based local evaluation	Clinicians and Practice Managers	Real World Use	Interviews	Thematic Analysis	Yes	**6**
Kukafka 2015	USA	Breast Cancer	Prognostic	Decision aid	Patients	Real World Use	Focus groups	Thematic Analysis (implied not stated)	Yes	**8**
McParland 2019	UK	DDx	DDx	Decision Support System	Clinicians	Real World Use	Focus groups	Thematic Analysis	No	**8**
Nieroda 2018	UK	Cancer	Clinical	Decision Support Tool	Clinicians and Patients	Workshop/Usability Testing	Focus groups, think aloud use of website, open	Thematic Analysis	No	**8**

For peer review only

							ended questionnaires			
Peiris 2009	Australia	CVD	Prognostic	Clinical Decision Support	Clinicians	Real World Use	Semi-structured interviews	Thematic Analysis	Yes	**8**
Porat 2017	UK	DDx	DDx	Decision Support System	Clinicians	Simulated Clinic	Semi-structured interviews	Thematic Analysis	No	**7**
Porat 2018	UK	Stroke	Prognostic	Decision Aid	Clinicians	Workshop/Usability Testing	Semi-structured interviews	Thematic Analysis	Yes	**3**
Porat 2019	UK	Stroke	Prognostic	Decision Aid	Clinicians, Patients, and other Stakeholders	Workshop/Usability testing	Focus groups, Interviews and usability evaluations.	Thematic Analysis	Yes	**5**
Richardson 2019	USA	Pharyngitis and Cough	Clinical	Decision Support Tools	Clinicians	Real World Use	Interview, video and audio of consultation	Constant Comparative Method	No	**8**
Saver 2014	USA	Diabetes complications	Prognostic	Outcome Model	Patients	Real World Use	Interviews	general inductive approach	Yes	**5**
Short 2003	UK	Stroke	Prognostic	Decision Support System	Clinicians	Simulated Clinic	Qualitative interview and questionnaire	Unclear	No	**7**
Short 2004	UK	Stroke	Prognostic	Decision Support System	Clinicians	Workshop/Usability testing	Interviews	Constant Comparative Method	Yes	**8**
Vail 2012	UK	Mental Health risk	Prognostic	Decision Support System	Clinicians	Workshop/Usability testing	Interviews, focus groups	Thematic Analysis	No	**8**

For peer review only

Voruganti 2015	Canada	Multiple	Prognostic	Risk Assessment Tools	Clinicians	Workshop/Usability testing	Focus Groups and usability testing	Grounded theory	No	**8**
Walker 2017	Australia	Colorectal Cancer	Prognostic	Risk Prediction Tool	Clinicians and Practice Managers	Simulated Clinic	Interviews, videoed simulated consultations analysed	Framework Analysis	No	**7**

Supplementary Table 2: Assessment of quality of studies with the CASP tool

Study ID	1. Was there a clear statement of the aims of the research?	2. Is a qualitative methodology appropriate?	3. Was the research design appropriate to address the aims of the research?	4. Was the recruitment strategy appropriate to the aims of the research?	5. Was the data collected in a way that addressed the research issue?	6. Has the relationship between researcher and participants been adequately considered?	7. Have ethical issues been taken into consideration?	8. Was the data analysis sufficiently rigorous?	9. Is there a clear statement of findings?	10. How valuable is the research?
Babb de Villiers 2019	Yes	Yes	No	No	Can't tell	No	No	No	No	No
Chiang 2015	Yes	Yes	Yes	Yes	Yes	Can't tell	Yes	Yes	Yes	Yes
Coe 2016	Yes	Yes	Can't tell	Yes	Yes	No	Yes	Yes	Yes	Yes
Collins 2013	Yes	Yes	Yes	Can't tell	Can't tell	Can't tell	Can't tell	Can't tell	Yes	Yes
Colombet 2003	Yes	Yes	Yes	Yes	Yes	Can't tell	Yes	Yes	Yes	Yes
Crawford 2010	Yes	Yes	Yes	Yes	Yes	Can't tell	Yes	Yes	Yes	Yes
Dikimitis 2015	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Du 2010	Yes	Yes	Yes	Can't tell	Can't tell	Can't tell	Can't tell	Can't tell	Can't tell	Can't tell
Eaton 2009	Yes	Yes	Can't tell	Can't tell	Yes	No	No	Can't tell	Yes	Yes
Harry 2019	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

For peer review only

Hawking 2019	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Jiwa 2006	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Can't tell	Can't tell
Keogh 2019	Yes	Yes	Yes	Yes	Yes	Can't tell	Yes	Yes	Yes	Yes
Kidney 2017	Yes	Yes	Yes	Can't tell	Yes	No	Yes	Can't tell	Yes	Yes
Kukafka 2015	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
McParland 2019	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Nieroda 2018	Yes	Yes	Yes	Yes	Yes	Can't tell	Yes	Yes	Yes	Yes
Peiris 2009	Yes	Yes	Yes	Yes	Yes	Can't tell	Yes	Yes	Yes	Yes
Porat 2017	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Porat 2018	No	Yes	Can't tell	Yes	Yes	No	No	No	Yes	Can't tell
Porat 2019	Yes	Yes	Yes	Yes	Yes	No	Can't tell	Can't tell	Yes	Can't tell
Richardson 2019	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Saver 2014	Yes	Yes	Can't tell	Yes	Can't tell	No	Yes	Yes	Yes	Can't tell
Short 2003	Yes	Yes	Yes	Can't tell	Yes	Can't tell	Yes	Yes	Yes	Yes
Short 2004	Yes	Yes	Can't tell	Yes	Yes	No	No	No	Yes	Yes
Vail 2012	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Voruganti 2015	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Walker 2017	Yes	Yes	Yes	Yes	Can't tell	Can't tell	Yes	Yes	Yes	Yes

Supplementary Table 3: Illustrative Quotes to reinforce Analytical Themes

Theme		Quote (all clinician comments, unless stated)	Study

For peer review only

Novel Risk	*Alerts to new risk*	*"It's just electronic highlight, makes you think even if you immediately dismiss it, at least that millisecond you've thought about it, and I think that is going to be useful at some point, but for how many people I don't know"*	Dikomitis 2015
		"Helpful to refresh your memory about things that are rare…we mainly stick to the phrase: 'common things are common"	Porat 2017
	Alerts to alternative risks	*"Yes, I must admit ovarian didn't come so high up, I had already fixated on the bowel stuff, which maybe I shouldn't have. This really said hey, consider ovarian as well. So again that was useful, yes"*	Chiang 2015
		"I think it widens your diagnosis, like the guy that had COPD and then ended up with AS. I would get to the diagnosis, but it would be longer without the tool."	Porat 2017
	Alerts to risk factor	*"the patient can see very clearly, [and say] 'Wow, smoking contributes a lot to cardiovascular risk, I do actually want to think about smoking cessation'"*	Voruganti 2015
		"I don't think it's (the eRAT) altered practice. I think it's kept some of the not so hard red flag symptoms in people's minds…"	Kidney 2017
Risk Refinement	*Orientation of risk (within guidelines/clinical pathways)*	*"Normally I'd get a few investigations, get the results back and then based on that say do we need to do something, or I refer this on based on that. But I guess if I have a calculator saying it's higher risk, it might prompt me to make a referral to a specialist a bit earlier."*	Chiang 2015
		"Think the biggest use I got, because you presented with very constitutional symptoms that a lot of things could present with, this was useful for me to think what's the good next step and what should I think of. So I was thinking in my mind more"	Chiang 2015
	More accurate risk, reduction of uncertainty	*For clinicians, the main perceived benefit was iPrevent's provision of accurate and reliable risk estimates for both ten year and lifetime risk, and standardized risk management advice for women*	Keogh 2019, Researcher comment
		"It is helpful in that it reminds me that the risk of stroke goes up substantially as you get older so your benefit (of aspirin) increases"	Short 2003
	Simplification of risk, reduction of uncertainty	*"Well I suppose you could think about the common comorbidities, so hypertension and stroke, AF (atrial fibrillation) and stroke, diabetes and stroke and you could think about not necessarily an algorithm but a sort of stepwise prioritisation about what you should think about in terms of the patient's total management, you know, which would be the most important area of focus?"*	Porat 2019

For peer review only

	Introduction of new risk knowledge	*"Other colleagues have said as well we're far more aware of thrombocytosis and increased platelets. We weren't aware that tended to increase the score for increased cancer risk"*	Dikomitis 2015
		"The learning aspect is very important, since this system is based on evidence and evidence can change"	Porat 2019
Communication	*Patient to Clinician Communication*	*"I think it's important to be informed and to get there as early as possible. To go and see the GP, and if there's a tool perhaps, you know, to help me do this, that may be of interest"*	Nieroda 2018 Patient Comment
	Risk focuses conversations	*"it usually stops a lot of the meandering dialogue that you'd otherwise engage in…. [initiate] that discussion in terms of what their risk factors are"*	Voruganti 2015
		"So having that piece of paper there gave you that conversation: 'Well now we've finished everything, let's look at this.'"	Peiris 2009
	Risk justifies action	*"But your heart is beating kinda fast, you've had a fever last night...the recommendation would be to get a chest x-ray today."*	Richardson 2019
		"OK, so our little risk calculator here is recommending that we would swab you for strep throat, and I agree with that."	Richardson 2019
	Risk justifies inaction	*"…there's a whole lot of people and this is your chance out of them of cancer…"*	Walker 2007
		"I think FRAX [is] useful because [it] can cut down on unnecessary investigations"	Voruganti 2015
	Improves understanding of risk	*"I think the biggest impact is that it changed the way I talked about what I was doing with patients, in that it made it a much more slick, neat package to describe the normal screening that you do for risk management. And so I felt it was easier to deliver some description of where they're at now. And from their point of view, I mean it's hard to know, but they seemed to understand that it was a multifactorial thing, rather than just being one of those single disease problems.... The thing that I don't really know, that I guess would be useful, is what they think when they walk out the door, what they actually understand of what I've said."*	Pieris 2009
		"With our Framingham tool you get…, a yellow bar. The average person's 10-year risk is…two little yellow bars, and your [the patient's] average risk is…maybe five little yellow bars'"	Voruganti 2015

For peer review only

		"I like this one [referring to the color bar].... I mean, everyone knows that red means danger, so if they're heading towards this one, it's a lot more visual, the impact...."	Peiris 2009
Autonomy	*Empowered by knowing personalised risk*	*"Anything that can be personalised or tailored, so you don't feel it's this off the shelf thing that you're being given, you know… you sit with your doctor and it's not just a case of giving out a leaflet, but actually let's have a look at your personal data"*	Porat 2019
		"I think the benefit is you're empowering yourself with information, you're possibly changing what you can change"	Keogh 2019, Patient comment
		"Anything that gets people to have a think about their health is a good thing, anything at all."	Walker 2007
Fear	*Direct fear of diagnosis*	*"Well I just kind of—a heart attack is a scariest thing to me. And I think it just kind of got me wondering and thinking and got me too nervous about the whole thing"*	Eaton 2009, Patient comment
		"And I thought, 'Wow! Patients coming with these tummy pains,' even if I'm thinking, 'She's got a 10% chance,' to actually think, 'Wow! Listen, you got a 38% chance of cancer.' General public are very—I would find that confronting and the general public have a shocking record at understanding risk."	Chiang 2015
	Disturbs normal practice	*"It's completely different to how I normally work, unsettling and confusing, I'm not quite sure where I am, to what extent I'm following a template and how I take history."*	Porat 2017
		"I personally don't think tools help me too much, interfere with my process of thought, would have done better without it, it was unnatural"	Porat 2017
	Removes focus and control of consultation	*"it might bring up a lot more other issues that they [patients] weren't originally aware of and the discussion might actually… be less directed"*	Voruganti 2015
		'Yeah. If I start saying to them, 'Oh god, your risk of having gastro-oesophageal cancer is one in three'y that seems way over the top, back pedalling from that, the consultation would've gone on for another 20 minutes or 30 minutes since I tried to explain to him, no. He doesn't have it'	Chiang 2015
	Lack of agency over risk	*The most commonly expressed concern was the potential for the risk assessment tool to miss something that was unknown and outside their control*	Kukafka 2015, Researcher Comment
	Medico-legal fear	*"Quite a few partners were worried about any medico-legal implications with*	Dikomitis 2015

For peer review only

		that. I think the one criticism was having the list of patient with PPVs. No one really liked that. That was worrying. We're worried about if patients knew that you had a list of them with the risk and you hadn't acted on it, what would be the implications? That was probably a point that put people off, really"	
		Some also felt that overriding a DDDSS would make them vulnerable to litigation	McParland 2019, Researcher Comment
Mistrust	*eRAT over-triggers*	*"Inevitably it will come up with a diagnosis of cancer. When the patient walks through the door . . . quite clearly, quickly, we can see this is not a serious diagnosis"*	McParland 2019
		"It gives information which, as it's blandly presented, you go, "How did you get that?…" I got a couple of people where I got a 20% number and you go, "Oh that's madness, that's not you," and often because it's based on single digit information…like a single blood pressure."	Peiris 2009
	eRAT misses things	*A consistent theme in our analysis was uncertainty among participants that all factors were accounted for in determining their breast cancer risk*	Kukafka 2015, Researcher comment
	eRAT not applicable to their patient	*In half of the patient visits, patients reported some piece of information, typically as a part of the history of present illness that raised a question for the coders of whether the tool was applicable to their clinical condition.*	Richardson 2019, Researcher comment
		they felt obliged to use the tool for the purpose of the study and their suspicion of cancer was unnaturally heightened	Chiang 2015
	Preference for diagnostic tests over risk scores	*"I would like to know if I have the gene mutation. So for me to say that being my mom had it [breast cancer] and take a pill and prevent it and not knowing if I have the gene doesn't make sense to me."*	Kukafka 2015
		"Colonoscopy was commonly referred to as the 'gold standard' for CRC screening. GPs were more likely to recommend colonoscopy if the patient wanted one or they had previously been recommended one by a gastroenterologist, even if the patient was at average risk."	Walker 2007, Researcher comment
	Preference for clinical judgement	*"… more experienced GPs were less likely to agree with the underlying concept and continued to maintain strong belief in their clinical intuition."*	Chiang 2015, Researcher comment
		"Patients don't follow flow charts, in my experience"	McParland 2019

For peer review only

		…that's just the tricky bit of saying [to the patient] that this is what's [sic] the recommendation but we don't always follow what recommendations are…	Walker 2007
	Patient preference for self knowledge	*"I think the model is incorrect because I know I am going to last longer than 10 years, because I do take care of myself"*	Saver 2014

For peer review only

ENTREQ checklist (Enhancing transparency in reporting the synthesis of qualitative research) *

No. Item	Guide questions/description	Reported on Page #
1. Aim	State the research question the synthesis addresses	Title & 3
2. Synthesis methodology	Identify the synthesis methodology or theoretical framework which underpins the synthesis, and describe the rationale for choice of methodology (e.g. meta-ethnography, thematic synthesis, critical interpretive synthesis, grounded theory synthesis, realist synthesis, meta-aggregation, meta-study, framework synthesis)	4
3. Approach to searching	Indicate whether the search was pre-planned (comprehensive search strategies to seek all available studies) or iterative (to seek all available concepts until they theoretical saturation is achieved)	4
4. Inclusion criteria	Specify the inclusion/exclusion criteria (e.g. in terms of population, language, year limits, type of publication, study type)	4
5. Data sources	Describe the information sources used (e.g. electronic databases (MEDLINE, EMBASE, CINAHL, psycINFO), grey literature databases (digital thesis, policy reports), relevant organisational websites, experts, information specialists, generic web searches (Google Scholar) hand searching, reference lists) and when the searches conducted; provide the rationale for using the data sources	4
6. Electronic Search strategy	Describe the literature search (e.g. provide electronic search strategies with population terms, clinical or health topic terms, experiential or social phenomena related terms, filters for qualitative research, and search limits)	Appendix
7. Study screening methods	Describe the process of study screening and sifting (e.g. title, abstract and full text review, number of independent reviewers who screened studies)	4
8. Study characteristics	Present the characteristics of the included studies (e.g. year of publication, country, population, number of participants, data collection, methodology, analysis, research questions)	Table 1
9. Study selection results	Identify the number of studies screened and provide reasons for study exclusion (e,g, for comprehensive searching, provide numbers of studies screened and reasons for exclusion indicated in a figure/flowchart; for iterative searching describe reasons for study exclusion and inclusion based on modifications to the research question and/or contribution to theory development)	4 and Figure 1
10. Rationale for appraisal	Describe the rationale and approach used to appraise the included studies or selected findings (e.g. assessment of conduct (validity and robustness), assessment of reporting (transparency), assessment of content and utility of the findings)	4

For Peer Review Only

11. Appraisal items	State the tools, frameworks and criteria used to appraise the studies or selected findings (e.g. Existing tools: CASP, QARI, COREQ, Mays and Pope [25]; reviewer developed tools; describe the domains assessed: research team, study design, data analysis and interpretations, reporting)	4
12.Appraisal process	Indicate whether the appraisal was conducted independently by more than one reviewer and if consensus was required	4
13.Appraisal results	Present results of the quality assessment and indicate which articles, if any, were weighted/excluded based on the assessment and give the rationale	Table 2
14. Data extraction	Indicate which sections of the primary studies were analysed and how were the data extracted from the primary studies? (e.g. all text under the headings "results /conclusions" were extracted electronically and entered into a computer software)	4
15. Software	State the computer software used, if any	4
16. Number of reviewers	Identify who was involved in coding and analysis	4
17. Coding	Describe the process for coding of data (e.g. line by line coding to search for concepts)	4
18. Study comparison	Describe how were comparisons made within and across studies (e.g. subsequent studies were coded into pre-existing concepts, and new concepts were created when deemed necessary)	4
19. Derivation of themes	Explain whether the process of deriving the themes or constructs was inductive or deductive	4
20. Quotations	Provide quotations from the primary studies to illustrate themes/constructs, and identify whether the quotations were participant quotations of the author's interpretation	Table 3
21. Synthesis output	Present rich, compelling and useful results that go beyond a summary of the primary studies (e.g. new interpretation, models of evidence, conceptual models, analytical framework, development of a new theory or construct)	5-7

* Reference: Tong A, Flemming K, McInnes E, Oliver SA, Craig J. Enhancing transparency in reporting the synthesis of qualitative research: ENTREQ. BMC Medical Research Methodology 2012, 12:181.

Search Strategy

#	Database	Search term	Results
1	Medline	("electronic risk assessment*").ti,ab	13
2	Medline	(electronic ADJ3 ("risk assessment*" OR RAT OR eRAT OR RATs OR eRATs OR diagnosis)).ti,ab	459

For peer review only

3	Medline	(comput* ADJ3 ("risk assessment*" OR RAT OR eRAT OR RATs OR eRATs OR diagnosis)).ti,ab	8879
4	Medline	"DECISION SUPPORT TECHNIQUES"/	19374
5	Medline	"DECISION SUPPORT SYSTEMS, CLINICAL"/ OR "DIAGNOSIS, COMPUTER-ASSISTED"/	29078
6	Medline	"RISK ASSESSMENT"/	249398
7	Medline	(electronic OR comput*).ti,ab	805823
8	Medline	(6 AND 7)	10675
9	Medline	("decision support").ti,ab	13295
10	Medline	("prediction tool*").ti,ab	3009
11	Medline	(qcancer).ti,ab	10
12	Medline	("risk prediction").ti,ab	8010
13	Medline	("computer guided consultation").ti,ab	2
24	Medline	(1 OR 2 OR 3 OR 4 OR 5 OR 8 OR 9 OR 10 OR 11 OR 12 OR 13)	82373
14	Medline	exp "GENERAL PRACTICE"/	73872
15	Medline	"PRIMARY HEALTH CARE"/	73929
16	Medline	"GENERAL PRACTITIONERS"/	7252
17	Medline	"PHYSICIANS, FAMILY"/	16158
18	Medline	("general practi*").ti,ab	38963
19	Medline	("primary health care").ti,ab	19769
20	Medline	("primary care").ti,ab	106822
21	Medline	(GP OR GPs).ti,ab	55107

For peer review only

22	Medline	("family physician*").ti,ab	14422
23	Medline	("family practice").ti,ab	7465
25	Medline	(14 OR 15 OR 16 OR 17 OR 18 OR 19 OR 20 OR 21 OR 22 OR 23)	274818
26	Medline	(24 AND 25)	3066
27	EMBASE	("electronic risk assessment*").ti,ab	30
28	EMBASE	(electronic ADJ4 ("risk assessment*" OR RAT OR eRAT OR RATs OR eRATs OR diagnosis)).ti,ab	811
29	EMBASE	(comput* ADJ4 ("risk assessment*" OR RAT OR eRAT OR RATs OR eRATs OR diagnosis)).ti,ab	12523
30	EMBASE	exp "DECISION SUPPORT SYSTEM"/ OR "COMPUTER ASSISTED DIAGNOSIS"/	61278
31	EMBASE	"RISK ASSESSMENT"/	534341
32	EMBASE	(electronic OR comput*).ti,ab	1181113
33	EMBASE	(31 AND 32)	24951
34	EMBASE	("decision support").ti,ab	17689
35	EMBASE	("prediction tool*").ti,ab	4544
36	EMBASE	(qcancer).ti,ab	14
37	EMBASE	("risk prediction").ti,ab	12919
38	EMBASE	("computer guided consultation").ti,ab	8
39	EMBASE	(27 OR 28 OR 29 OR 30 OR 33 OR 34 OR 35 OR 36 OR 37 OR 38)	117361
40	EMBASE	"GENERAL PRACTICE"/	76245
41	EMBASE	exp "PRIMARY HEALTH CARE"/	162467

For peer review only

42	EMBASE	"GENERAL PRACTITIONER"/	94915
43	EMBASE	("general practi*").ti,ab	96563
44	EMBASE	("primary health care").ti,ab	23923
45	EMBASE	("primary care").ti,ab	146773
46	EMBASE	(GP OR GPs).ti,ab	79393
47	EMBASE	("family physician*").ti,ab	17848
48	EMBASE	("family practice").ti,ab	7988
49	EMBASE	(40 OR 41 OR 42 OR 43 OR 44 OR 45 OR 46 OR 47 OR 48)	415195
50	EMBASE	(39 AND 49)	4595
51	CINAHL	("electronic risk assessment*").ti,ab	7
52	CINAHL	(electronic ADJ4 ("risk assessment*" OR RAT OR eRAT OR RATs OR eRATs OR diagnosis)).ti,ab	279
53	CINAHL	(comput* ADJ4 ("risk assessment*" OR RAT OR eRAT OR RATs OR eRATs OR diagnosis)).ti,ab	2066
54	CINAHL	"DECISION SUPPORT SYSTEMS, CLINICAL"/	4554
55	CINAHL	exp "DECISION SUPPORT TECHNIQUES"/	9645
56	CINAHL	"DIAGNOSIS, COMPUTER ASSISTED"/	4031
57	CINAHL	"RISK ASSESSMENT"/	87369
58	CINAHL	(electronic OR comput*).ti,ab	184706
59	CINAHL	(57 AND 58)	4195
60	CINAHL	("decision support").ti,ab	5671
61	CINAHL	("prediction tool*").ti,ab	678

For peer review only

62	CINAHL	(qcancer).ti,ab	8
63	CINAHL	("risk prediction").ti,ab	2809
64	CINAHL	("computer guided consultation").ti,ab	1
65	CINAHL	(51 OR 52 OR 53 OR 54 OR 55 OR 56 OR 59 OR 60 OR 61 OR 62 OR 63 OR 64)	29549
66	CINAHL	"FAMILY PRACTICE"/	22717
67	CINAHL	"PHYSICIANS, FAMILY"/	17517
68	CINAHL	"PRIMARY HEALTH CARE"/	56524
69	CINAHL	("general practi*").ti,ab	26608
70	CINAHL	("primary health care").ti,ab	10032
71	CINAHL	(GP OR GPs).ti,ab	18016
72	CINAHL	("family physician*").ti,ab	5248
73	CINAHL	("family practice").ti,ab	1902
74	CINAHL	(66 OR 67 OR 68 OR 69 OR 70 OR 71 OR 72 OR 73)	111915
75	CINAHL	(65 AND 74)	1406

Web of Science Search

(“electronic risk assessment” OR electronic NEAR/3 ("risk assessment*" OR RAT OR eRAT OR RATs OR eRATs OR diagnosis) OR comput* NEAR/3 ("risk assessment*" OR RAT OR eRAT OR RATs OR eRATs OR diagnosis) OR “decision support” OR "prediction tool*" OR qcancer OR “risk prediction” OR "computer guided consultation") AND (“General practi*” OR “Primary Health Care” OR “Family Physician*” OR “Primary Care” OR GP OR GPs OR “Family Practice”)

For peer review only

BMJ Open

How do electronic risk assessment tools affect the communication and understanding of diagnostic uncertainty in the primary care consultation? A systematic review and thematic synthesis

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-060101.R2
Article Type:	Original research
Date Submitted by the Author:	09-Jun-2022
Complete List of Authors:	Burns, Alex; University of Exeter, College of Medicine and Health; PenARC, Donnelly, Brian; University of Exeter Medical School Feyi-Waboso, Joshua; University of Exeter Medical School Shephard, Elizabeth; University of Exeter, College of Medicine and Health Calitri, Raff; University of Exeter, College of Medicine and Health Tarrant, Mark; University of Exeter, College of Medicine and Health Dean, Sarah; PenARC; University of Exeter, College of Medicine and Health
Primary Subject Heading:	General practice / Family practice
Secondary Subject Heading:	Diagnostics, Qualitative research
Keywords:	PRIMARY CARE, QUALITATIVE RESEARCH, Risk management < HEALTH SERVICES ADMINISTRATION & MANAGEMENT

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

How do electronic risk assessment tools affect the communication and understanding of diagnostic uncertainty in the primary care consultation? A systematic review and thematic synthesis

Dr Alex Burns MBBS, MGHP, BA(cantab). GP Partner, Threespires Medical Practice, Truro. PhD Student, University of Exeter, College of Medicine and Health. ORCID 0000-0002-9931-9198

Brian Donnelly. Medical Student, University of Exeter Medical School

Joshua Feyi-Waboso. Medical Student, University of Exeter Medical School. ORCID 0000-0002-5417-4298

Dr Elizabeth Shephard PhD, MSc, BSc. Research Fellow, University of Exeter, College of Medicine and Health. ORCID 0000-0002-3610-3680

Dr Raff Calitri PhD, MSc, BSc. Research Fellow, University of Exeter, College of Medicine and Health. ORCID 0000-0003-0889-4670

Dr Mark Tarrant PhD, BA(hons). Associate Professor, University of Exeter, College of Medicine and Health. ORCID 0000-0002-7194-1428

Professor Sarah Dean, PhD, MSc, BSc. Professor, University of Exeter, College of Medicine and Health. ORCID 0000-0002-3682-5149

corresponding author: Dr Alex Burns, ab1265@exeter.ac.uk

For peer review only

How do electronic risk assessment tools affect the communication and understanding of diagnostic uncertainty in the primary care consultation? A systematic review and thematic synthesis

Alex Burns, Brian Donnelly, Joshua Feyi-Waboso, Elizabeth Shephard, Raff Calitri, Mark Tarrant and Sarah Dean

Abstract

Objectives: To conduct a systematic review and synthesise qualitative research of electronic risk assessment tools (eRATs) in primary care, examining how they affect the communication and understanding of diagnostic risk and uncertainty. eRATs are computer-based algorithms designed to help clinicians avoid missing important diagnoses, pick up possible symptoms early and facilitate shared decision making.

Design: Systematic search, using predefined criteria of the published literature and synthesis of the qualitative data, using Thematic Synthesis. Database searches on 27th November 2019 were of MEDLINE, EMBASE, CINAHL and Web of Science, and a secondary search of the references of included articles. Included studies were those involving electronic risk assessment or decision support, pertaining to diagnosis in primary care, where qualitative data were presented. Non-empirical studies and non-English language studies were excluded. 5971 unique studies were identified of which 441 underwent full text review. 26 studies were included for data extraction. A further 2 were found from citation searches. Quality appraisal was via the CASP tool. Data extraction was via line by line coding. A Thematic Synthesis was performed.

Setting: Primary Care

Results: Electronic risk assessment tools included differential diagnosis suggestion tools, tools which produce a future risk of disease development or recurrence, or calculate a risk of current undiagnosed disease. Analytical themes were developed to describe separate aspects of the clinical consultation where risk and uncertainty are both central and altered via the use of an eRAT: “Novel risk”, “Risk refinement”, “Autonomy”,” Communication”, “Fear” and “Mistrust”.

Conclusion: eRATs may improve the understanding and communication of risk in the primary care consultation. The themes of “Fear” and “Mistrust” could represent potential challenges eRATs may create.

Registered review with Prospero, CRD219446

Strengths and Limitations:

- This is the first review to address the impact of eRATs in the primary care consultation
- There is inconsistency in nomenclature surrounding these tools
- There is a lack of data from patients in the primary studies
- There are likely biases in the original studies.

For peer review only

Introduction

Diagnosis in primary care is complex and challenging. The main input, a patient's history and examination, can be variable, imperfect and subjective. Pre-test probability is lower than other health-care settings[1], [2]. GPs are able to process and communicate risk and accept uncertainty in diagnosis:[3] Doing this in a way which allows a shared understanding of medicine with a patient is one of the seminal challenges of primary care.[4]

Clinical Decision Support Systems are designed to "enhance medical decisions with targeted clinical knowledge, patient information, and other health information"[5]. "Electronic Risk Assessment Tools" (hereafter referred to as eRATs) are decision support systems which help with diagnosis. They are software where information about a patient is processed electronically to produce an output pertinent to a diagnosis, either immediately or in the future. The output can be a quantitative risk of a specific diagnosis, or a more qualitative output: prompting the clinician to consider a particular diagnosis or list of possible diagnoses. eRATs can enable extraction of data previously inputted into primary care IT systems, and can prompt users to input missing data. They can incorporate "pop ups" during a consultation to avoid missing important diagnoses.[6], [7] They have a range of stated aims: making diagnosis more evidence based and objective; avoiding missed diagnoses; or facilitating shared decision making.

eRATs tend to target diagnoses which are both serious and benefit from early intervention,[6], [7] an obvious example being cancer. To achieve adequate sensitivity these algorithms tend towards a low positive predictive value.[6], [7] As eRATs use more complex calculations, invisible to the clinician, and insert themselves into the clinician consultation more proactively, they may come to represent a "third actor" in the consultation, alongside the patient and clinician.[8] They have the potential to change the consultation dynamic and diagnostic thinking process. Although there have been systematic reviews conducted on the effectiveness of eRATs[9], [10], there have been no systematic reviews using qualitative methods examining risk and uncertainty in their use.

Diagnosis in primary care tends to be the outcome of clinical reasoning, which can be considered using the theories of Situativity:[11] that diagnosis will depend on other actors (as well as the clinician) present during the consultation, and their interaction with multiple environmental inputs; in our case eRATs. Within this clinical situation we are interested in the communication of risk, which is the best proxy we have of the understanding of risk. The Theory of Communicative Rationality [12] may help our understanding - communication can be emancipative: an attempt to achieve mutual understanding (as in shared decision making); or strategic, where the intention of the communication is to achieve a particular aim of an individual, such as calculative manipulation of others.

Aim

This study aims to synthesise qualitative research on eRATs in the primary care consultation, examining how they affect the communication and understanding of the risk of a particular disease, and uncertainty when used making diagnoses.

Methods

For peer review only

Study Selection

Searches of MEDLINE, EMBASE, CINAHL and Web of Science were performed. The search strategy[13] targeted studies involving a primary care population, where risk assessment or decision support tools are used for, where qualitative data are reported regarding risk or uncertainty. Non-empirical studies and non-English language studies were excluded, as well as studies outside of primary care, or where qualitative data were not presented. Using Covidence software[14], three authors (AB, JF-W and BD) independently screened titles and abstracts, and then full text for relevance. Disputes were resolved on discussion between authors. The initial 5971 unique studies that were identified reduced to 441 on title/abstract review. These underwent full text review, which yielded 26 individual studies, and one systematic review[9] (Figure 1). These were hand searched for further citations, which yielded 2 additional studies. Key eRATs embedded into System1, a mainstream UK IT system, were identified and the developers emailed (where qualitative studies had not already been identified) to see if there were unpublished data available. This did not identify further studies. Thus 28 independent studies were included for data extraction (supplementary table1).

Quality appraisal Data Extraction

The 28 studies were assessed for quality (AB, ES and SD) using the CASP critical appraisal checklist[15]. The CASP checklist was designed to be used as educational pedagogic tool and not as a scoring tool for systematic reviews. A point was assigned if the study met one of the 10 statements making up the CASP checklist. Discrepancies were resolved on discussion between the authors (supplementary table 2). None were excluded from analysis due to low quality[16]. Data extraction was performed by AB. Quotes and researcher statements regarding risk and uncertainty were extracted electronically and coded into NVivo software.[17]

Thematic Synthesis

Thematic Synthesis was chosen as it is based in Scientific Realism[18]: the purpose of this review is to examine how clinicians and patients understanding of risk and uncertainty change with the use of eRATs. For the results to be broadly applicable, the underlying assumption is that the understanding of risk, and this communication is consistent, or consistent within a range.

Descriptive codes were developed by AB. Codes were described in the initial studies coded, and subsequent studies coded into these concepts, with new descriptive codes being created, refined or ordered hierarchically when deemed necessary. These descriptive codes were discussed with other authors (SD, ES, RC and MT). Thematic Synthesis involves the “translation of concepts from one study to another”[19]. The ten studies with analytical findings pertaining to risk or uncertainty were examined. The developed descriptive codes were applied to each to explore new coding approaches. The descriptive codes were then developed inductively into analytical themes describing separate aspects of the clinical consultation where risk and uncertainty is both central to, and altered via, the use of an eRAT. Selective quotes are used to illustrate the richness of data and to provide support for these themes.

Patient and public involvement

Patients and the public were not involved in the review

Research ethics approval

This study did not receive or require ethics approval, as it does not involve human or animal participants

Results

Three conceptually different forms of eRATs were identified: those which produced a risk of an immediate, or "clinical" diagnosis, (e.g., the risk of a current bacterial tonsillitis); those which produced a risk of a future or "prognostic" diagnosis, (e.g., the risk of cardiovascular disease in the next decade); and those which produced a list of potential alternative diagnoses, or "differential diagnosis generators". This is an important distinction because it gets to the heart of what diagnosis is: separating the affected from the non-affected. This means defining illness, and the concept of current illness is distinct to the threat of future illness.[20] It has further implications in that both clinician and patient may be able to change, and perhaps have responsibility for, "prognostic" risk. It was decided that the three conceptual eRATs had aspects which were similar enough to attempt a synthesis: to see if there were findings which applied across the forms of eRATs, and that could help answer the research question.[21]

Except for a single study,[22] those concerning eRATs for clinical diagnosis collected data from clinicians rather than patients. Studies of prognostic eRATs were more orientated towards the communication of risk to patients. The studies examining differential diagnosis generators had data from both clinicians and patients.

Included studies had publication dates ranging from 2003 to 2019. [22]–[49] There was a broad range of targeted diagnoses, as well as data collection and analytical methods. There was also a range of nomenclature used to describe the eRAT being used. Details are available in supplementary table 1. Only ten of the studies reported themes or findings pertaining to risk or uncertainty in their analysis.[27], [33]–[35], [37]–[40], [42], [49]

Analytical Themes:

Six analytical themes are presented below with supportive quotes (further supportive quotes are presented in supplementary table 3). Quotes are from clinicians, unless specified otherwise.

"Novel risk". A new risk is introduced into the consultation, either as an unconsidered diagnosis or an unexpected high risk.[24], [26], [28], [33], [46], [47] It brings the risk to the forefront of the consultation and ensures that it is adequately considered.

"It's just electronic highlight, makes you think even if you immediately dismiss it, at least that millisecond you've thought about it, and I think that is going to be useful at some point, but for how many people I don't know."[26]

"I think it's important to be informed and to get there as early as possible. To go and see the GP, and if there's a tool perhaps, you know, to help me do this, that may be of interest." Patient Comment[22]

The eRAT can raise the risk of alternative diagnoses:

For peer review only

"Yes, I must admit ovarian didn't come so high up, I had already fixated on the bowel stuff, which maybe I shouldn't have. This really said hey, consider ovarian as well. So again that was useful, yes."[24]

This theme arose mainly from studies regarding clinical diagnosis, and differential diagnosis generators, and arose mainly from data regarding clinicians views.

"Risk refinement". In this theme, the risk is processed into a more accurate assessment. It was generated from studies concerning eRATs for prognostic and clinical diagnosis, and mainly from clinicians.

It enables clinicians: "t*o take complex, competing information and bring it together".*[36]

This refinement of risk serves several purposes. It can give an objective single point estimate of risk.[28], [36] it can take complex risk factors and combine them into a single risk output,[37] and it can orientate the clinician towards guidelines and clinical pathways, triggering them to add missing information to the eRAT,[39] suggesting the next diagnostic tests which are appropriate[24] to reduce uncertainty and improve diagnostic accuracy.

"Communication". eRATs can assist clinicians to communicate risk to patients, and in patient-facing eRATs, to help patients communicate with their clinicians[22]. They can provide a focus point for a consultation.[39], [46] Clinicians can use the eRAT output to justify their next action, be this referral or further testing particularly in eRATs for clinical diagnosis.[24], [45] Where the risk output is low they can justify inaction.[24], [31] The eRAT acts as a tool to communicate this:

"OK, so our little risk calculator here is recommending that we would swab you for strep throat, and I agree with that."[45]

"I think FRAX [is] useful because [it] can cut down on unnecessary investigations."[46]

Prognostic eRATs, can emphasise the level of risk, with a preference for colour displays or other qualitative methods of showing risk.[39], [46] eRAT outputs can be used as a motivator to take treatments or change behaviour.[38]

"…you're possibly changing what you can change" Patient comment[28]

"Autonomy". The eRAT output is a personalised risk pertaining to a patient's health, enabling them to make informed decisions. [28], [37] The eRAT was an enabler of this as it provided a personalised risk and prompted patients to think about it.

"Anything that gets people to have a think about their health is a good thing, anything at all."[31]

The fact that the eRAT has an output which not generic, but personalised to the patient's data, was perceived to be an important component of this benefit amongst clinicians and patients across the range of eRATs examined in this review.

"Fear". The eRAT could create anxiety or disquiet in both patient and clinician. This can occur at several levels. The first is the fear of a serious diagnosis from the eRAT. Patients have understandable concern about their own health and welfare:[34]

"If I knew I was from a high risk category, and I'm pretty sure I'm not, I reckon I would find that so

For peer review only

stressful. I would just worry every time there was a pimple", Patient comment[28]

clinicians share this fear, with an overlay of concern about poor understanding of risk among patients:

"And I thought, 'Wow! Patients coming with these tummy pains,' even if I'm thinking, 'She's got a 10% chance,' to actually think, 'Wow! Listen, you got a 38% chance of cancer.' General public are very—I would find that confronting and the general public have a shocking record at understanding risk."[24]

A second fear from clinicians regarding differential diagnosis generators, and clinical diagnosis eRATs, was of the tool interrupting how they normally think about diagnosis.[47], [48] The eRAT could distort how they handled diagnostic risk and unwanted risk suggestions made clinicians uncertain. There was a fear that the risk outputs could lead to a loss of control of the consultation, as the uncertainties raised by the eRATs override other agendas.[24], [46]

Finally, there were fears expressed regarding the medico-legal consequences of eRATs. Clinicians expressed concern that any eRAT output, if not acted on, could act as evidence against them[26], [48] and make them vulnerable to litigation. Here though, there was a negative case in the data, with one clinician reporting the eRAT was reassuring: that it acted as a "backstop", reducing fear of missing a diagnosis.

"So although it can cause irritation, the flip side of that is you can relax a little bit more and that you don't have to remember absolutely everything (…) You know, how to sort the wheat from the chaff. So although people might think it's irritating, it can be reassuring."[26]

"Mistrust". There was a theme of mistrust in the risk output of the eRAT. Some clinicians felt that the eRAT algorithm was flawed: because the trigger was set at an incorrect level of risk;[25], [48] that it relied on a single parameter, and changes to that parameter could disproportionately change the output;[39] that the inputs themselves were out of date or flawed in some way; or that the outputs were outside the control of patient and clinician. There was then a suspicion that the eRAT could be either under or over reporting a risk.

"Inevitably it will come up with a diagnosis of cancer. When the patient walks through the door... quite clearly, quickly, we can see this is not a serious diagnosis"[48]

Although clinicians could use the eRAT output to justify not performing further investigations, this tended to be in high risk situations where further testing was unnecessary before diagnosis and action. In situations where there was a low risk output from the eRAT, both clinicians and patients preferred diagnostic tests over the output of an eRAT.[31], [49] Here, a diagnostic test would be ordered despite the eRAT suggesting a risk low enough that testing was unnecessary. The test was trusted and the eRAT discounted.

There was faith in personal judgement over the eRAT. Patients preferentially trusting their knowledge of their own bodies,[28], [42] clinicians trusting their clinical acumen or gut instinct.[24], [31], [40], [48] Over-ruling an eRAT with clinical acumen was seen as good medicine:

"Patients don't follow flow charts, in my experience."[48]

Discussion

For peer review only

Summary: This review found that eRATs can influence diagnosis, and the communication of its risks and uncertainties in differing ways. Situativity[11] includes the idea of Distributed Cognition[50], where reasoning is spread between several actors or tools. An eRAT introducing "Novel" risk in a consultation could be seen as an example of this, and could have the potential to overcome the cognitive errors of misdiagnosis (such as premature closure).[51] Shared Decision Making[52] could be seen as a combination of the themes of "Communication", "Autonomy" and "Risk Refinement", helping realise the Emancipative Communication as described by Habermas[12]. In addition, the "Risk Refinement" function of an eRAT has the potential to increase the consistency of diagnostic decision making, which might improve adherence to guidelines, and play a part in cost containment.[5] These more positive themes could be undermined by those of "Mistrust" and "Fear", where the risk outputs of the eRAT are disregarded, or seen as something that could challenge or disturb the communication of diagnostic risk.

Strengths and limitations: This review is the first to address how eRATs affect understanding and communication of risk and uncertainty in the primary care consultation. The broad scope of this review means a range of eRATs have been synthesised. Some themes presented derive a greater part of their data from eRATs of a particular conceptual design ("clinical", "prognostic" or "differential diagnosis", as described in the results section), which could challenge their transferability across all eRATs. This review was intended to look at the wider clinician and patient interaction (in terms of understanding and communication) with eRATs and the findings are broadly transferable, which supports the analytical approach taken.

Another challenge concerns the nomenclature surrounding these tools: we used search terms to capture all studies where information was processed electronically to produce an output pertinent to a diagnosis, either immediately or in the future: here we have called them eRATs. The language used by the originators is often different and has changed over time. eRATs can also be referred to by other names, such as Diagnostic Decision Support Systems, Electronic Screening Tools and Risk Prediction Tools. The review may have missed interventions which, if examined, might be an eRAT by our definition.

There was limited data from patient's views regarding eRATs: the majority of the qualitative data regarding risk and uncertainty came from clinicians in the review's studies, and data from patients was less conceptually rich. If sharing diagnostic decisions with patients is to occur[53] more research is needed into the patient's perspective. There were limited findings regarding uncertainty, compared to those concerning risk. Uncertainty in medicine is an under-researched area [3], [54]–[56] thus eRATs are influencing aspects of the consultation which are already poorly understood.

Thematic synthesis includes "translation" of the review findings into the primary studies' thematic frameworks. This involves "*the process of taking concepts from one study and recognising the same concepts in another study*". [19] However, understanding and communication of risk and uncertainty was included in the analytical findings of only ten of the primary studies, limiting this aspect of the synthesis. Single author data extraction is also a limitation, as the findings may be more prone to author bias.

Comparison with existing literature: There are studies supporting the use of alerts regarding prescribing errors [5], [57], [58], and the use of similar alerts to avoid missed diagnoses is a logical step. However, this introduction of novel risk into the consultation deserves a sound evidence base

For peer review only

which is currently missing. Mistrust of decision support systems is a recognised issue[59] and may be justified: there is limited evidence supporting eRATs in terms of improving survival and patient experience.[12,13] The role of clinical judgement in the mistrust of eRATs is a tension which has been described in other areas of evidence based medicine.[60], [61] Part of clinical judgement is a clinicians' gut feelings. The role of gut feelings has been studied in regards to serious diagnoses [62], [63] although these focused on gut feelings triggering action despite a lack of (what might be considered) objective risk, rather than decisions not to act despite risk information: the latter is likely more common in the low prevalence setting of primary care, and this area warrants further study.

The role of fear in diagnosis, although well recognised,[64] is not well researched. A more mature literature exists for fear appeals in public health messaging. Witte et al[65] has examined the Extended Parallel Process Model[66], where fear appeals produce two competing responses which interfere with each other: "danger control", where the response is to avoid the threat producing the fear, and "fear control" where the fear itself is avoided, and a health recommendation defensively resisted. Thus, fear generated by eRATs may lead to the avoidance of the eRAT, rather than any intended outcome. Fear is generally a concerning response: the systemic harms of fear on GP retention have been described[67]. Although clinical uncertainty is an inevitable part of the diagnostic process, when uncertainty becomes fear, it is likely to lead to worse overall decisions, and disengage clinicians from use of eRATs.

Implications for future research and practice: Future research is needed into the understanding of diagnostic risk and uncertainty, from both patient and clinician's view points, how they differ and how eRATs influence the resulting communication. More evidence is needed examining how eRATs influence a clinicians clinical reasoning, particularly when the eRAT challenges clinical judgement. This review may provide a framework through which a clinician and their patient may consider their own interactions with eRATs. The approach needs to be a nuanced balance: to value a clinician's judgement; to be aware of the strengths and limitations of eRATs and what they bring to the consultation, whilst minimising the fear they generate for both patient and clinician.

Contributors AB: protocol, literature search, citation and full text screening, quality appraisal, data analysis, article. JF-W & BD: citation and full text screening. ES and SD: quality appraisal. ES, RC, MT and SD supervision at all stages.

Competing interests: There are no conflicts of interest declared by any author.

Funding: AB is a PhD candidate in a post funded by the University of Exeter (PhD Studentship Ref 3521) in conjunction with the ERICA trial (http://www.theericatrial.co.uk), which is funded by the Dennis and Mirelle Gillings Foundation, and receives support from the charities Cancer Research UK and Macmillan.

Data sharing statement: No additional data available

Figure 1 caption: Flow Diagram for systematic searches of databases and registers

References

1. Buntinx F, Mant D, Van Den Bruel A, et al. Dealing with low-incidence serious diseases in

general practice. Br J Gen Pract. 2011 Jan; 61(582):43–6.
2. Hamilton W. Cancer diagnosis in primary care. Br J Gen Pract. 2010 Feb;60(571):121–7.
3. O'Riordan M, Miguel J, Akturk Z, et al. Dealing with uncertainty in general practice: an essential skill for the general practitioner. Qual Prim Care. 2011; 19(3):175–81.
4. Maskrey N, Gordon A. Shared understanding with patients. JAMA Intern Med. 2017 Sep 1;177(9):1247–8.
5. Sutton RT, Pincock D, Baumgart DC, et al. An overview of clinical decision support systems: benefits, risks, and strategies for success. NPJ Digit Med. 2020 Feb 6;3:17.
6. NHS forced to admit sepsis guidance is "difficult" as GPs switch off alerts - Pulse Today. *Pulse Magazine* 2017. https://www.pulsetoday.co.uk/news/clinical-areas/infectious-diseases/nhs-forced-to-admit-sepsis-guidance-is-difficult-as-gps-switch-off-alerts/
7. Hamilton, W. Electronic Risk Assessment for Cancer for Patients in General Practice (ISRCTN22560297). *ISRCTN Registry* https://www.isrctn.com/ISRCTN22560297.
8. Shachak, A. & Reis, S. The Computer Assisted Patient Consultation. in *Clinical Technologies* 160–171 (IGI Global, 2011). doi:10.4018/978-1-60960-561-2.
9. Chima S, Reece JC, Milley K, et al. Decision support tools to improve cancer diagnostic decision making in primary care: a systematic review. Br J Gen Pract . 2019; 69(689):e809–18.
10. Riches N, Panagioti M, Alam R, et al. The effectiveness of electronic differential diagnoses (DDX) generators: A systematic review and meta-analysis. PLoS One. 2016 Mar 1;11(3):e0148991.
11. Durning, S. J., Artino, A. R. J., Schuwirth, L. & van der Vleuten, C. Clarifying Assumptions to Enhance Our Understanding and Assessment of Clinical Reasoning. Academic Medicine. 2013, 88.
12. Habermas, J. The Theory of Communicative Action,1984, Beacon Press.
13. Burns A, Donnelly B, Feyi-Waboso J, Dean S. How do electronic risk assessment tools affect the communication and understanding of diagnostic uncertainty in the primary care consultation. Do they alter how clinicians communicate risk and uncertainty? Is the understanding of uncertainty altered for either clinician or patient? . PROSPERO: International prospective register of systematic reviews. 219446. 2020. Available from: https://www.crd.york.ac.uk/prospero/display_record.php?RecordID=219446
14. Veritas Health Innovation, Melbourne A. Covidence systematic review software . Available from: www.covidence.org
15. Critical Appraisal Skills Program. CASP Qualitative Studies Checklist . 2018. Available from: https://casp-uk.b-cdn.net/
16. Garside R. Should we appraise the quality of qualitative research reports for systematic reviews, and if so, how? Innov Eur J Soc Sci Res 2013, 27:67–79.
17. International QSR. NVivo qualitative data analysis software. 2020.
18. Barnett-Page, E. & Thomas, J. Methods for the synthesis of qualitative research: A critical review. BMC Medical Research Methodology 2009, 9, 1–11.
19. Thomas, J. & Harden, A. Methods for the thematic synthesis of qualitative research in systematic reviews. BMC Medical Research Methodology 2008, 8:45
20. Coggon, D., Barker, D. & Rose, G. Epidemiology for the Uninitiated, 2003. Wiley.
21. Weir MC, Grimshaw JM, Mayhew A, et al. Decisions about lumping vs. splitting of the scope of systematic reviews of complex interventions are not well justified: A case study in systematic reviews of health care professional reminders. J Clin Epidemiol. 2012;65(7):756–63.
22. Nieroda ME, Lophatananon A, McMillan B, et al. Online Decision Support Tool for Personalized Cancer Symptom Checking in the Community (REACT): Acceptability, Feasibility, and Usability Study. JMIR cancer. 2018;4(2):e10073.

For Peer Review Only

23. Babb De Villiers C., Antoniou AC, Easton D, et al. Development of cancer risk prediction models into tools for use in clinical settings. Eur J Hum Genet. 2019;26 (Supplement 1):789.
24. Chiang PP-C, Glance D, Walker J, et al. Implementing a QCancer risk tool into general practice consultations: an exploratory study using simulated consultations with Australian general practitioners. Br J Cancer. 2015;112 Suppl:S77-83.
25. Jiwa M, Skinner P, Coker, AO, et al. Implementing referral guidelines: lessons from a negative outcome cluster randomised factorial trial in general practice. BMC Fam Pr. 2006;7(PG-65):65.
26. Dikomitis L, Green T. Embedding electronic decision-support tools for suspected cancer in primary care: a qualitative study of GPs' experiences. Prim Health Care Res Dev. 2015;16(6):548–55.
27. Kidney E, Greenfield S, Berkman L, et al. Cancer suspicion in general practice, urgent referral, and time to diagnosis: a population-based GP survey nested within a feasibility study using information technology to flag-up patients with symptoms of colorectal cancer. BJGP Open. 2017;1(3 PG-bjgpopen17X101109)
28. Keogh LA, Steel E, Weideman P, et al. Consumer and clinician perspectives on personalising breast cancer prevention information. Breast. 2019;43:39–47.
29. Collins I, Keogh L, Steel E, et al. Development of a tailored, computerized, breast cancer risk assessment and decision support tool: What do clinicians want? J Clin Oncol. 2013;31(15 SUPPL. 1).
30. Coe AM, Ueng W, Vargas JM, et al. Usability Testing of a Web-Based Decision Aid for Breast Cancer Risk Assessment Among Multi-Ethnic Women. AMIA. Annu Symp proceedings AMIA Symp. 2017 Feb 10:411–20.
31. Walker J, Bickerstaffe A, Hewabandu N, et al. A novel colorectal cancer risk prediction tool (CRISP): From inception to implementation. Asia Pac J Clin Oncol . 2017;13(Supplement 4):111.
32. Du X, Li S, Bhattacharyya O, et al. Identifying barriers toward implementing absolute cardiovascular risk assessment tools in primary care setting. Circulation. 2010;122(2):e363.
33. Hawkin M, Timmis A, Wilkins F, et al. Improving cardiovascular disease risk communication in NHS Health Checks: a qualitative study. BMJ Open. 2019;9(8):e026058.
34. Eaton CB, Parker DR, Craft J, et al. Using e-health to improve cholesterol management in primary care practice. J Med Pract Manage. 2009;24(4):224–30.
35. Short D, Frischer M. Barriers to the adoption of computerised decision support systems in general practice consultations: A qualitative study of GPs' perspectives. Int J Med Inform. 2004;73(4):357–62.
36. Short D, Frischer M, Bashford J. The development and evaluation of a computerised decision support system for primary care based upon "patient profile decision analysis". Inform Prim Care. 2003;11(4):195–202.
37. Porat T, Marshall IJ, Sadler E, et al. Collaborative design of a decision aid for stroke survivors with multimorbidity: A qualitative study in the UK engaging key stakeholders. BMJ Open. 2019;9(8):e030385.
38. Porat T, Liao Z, Curcin V, et al. Engaging Stakeholders in the Design and Usability Evaluation of a Decision Aid to Improve Secondary Stroke Prevention. Stud Health Technol Inform. 2018;247:765–9.
39. Peiris DP, Joshi R, Webster RJ, et al. An electronic clinical decision support tool to assist primary care providers in cardiovascular disease risk management: development and mixed methods evaluation. J Med Internet Res. 2009;11(4):e51.
40. Colombet I, Dart T, Leneveut L, et al. A computer decision aid for medical prevention: a pilot qualitative study of the Personalized Estimate of Risks (EsPeR) system. BMC Med Inform Decis Mak. 2003;3:13.
41. Harry ML, Truitt AR, Saman DM, et al. Barriers and facilitators to implementing cancer

For peer review only

prevention clinical decision support in primary care: a qualitative study. BMC Health Serv Res. 2019;19(1):534.
42. Saver B, Mazor K, Hargraves J, et al. Inaccurate risk perceptions and individualized risk estimates by patients with type 2 diabetes. J Am Board Fam Med. 2014;27(4 PG-510–9):510–9.
43. Crawford F, Bekker HL, Young M, et al. General practitioners' and nurses' experiences of using computerised decision support in screening for diabetic foot disease: implementing Scottish Clinical Information - Diabetes Care in routine clinical practice. Inform Prim Care. 2010;18(4):259–68.
44. Vail L, Adams A, Gilbert E, et al. Investigating mental health risk assessment in primary care and the potential role of a structured decision support tool, GRiST. Ment Health Fam Med. 2012;9(1):57–67. 08183CFB6CC4BDEDC0D64F5
45. Richardson S, Feldstein D, McGinn T, et al. Live Usability Testing of Two Complex Clinical Decision Support Tools: Observational Study. JMIR Hum Factors. 2019 Apr 15;6(2):e12471.
46. Voruganti T., O'Brien M, Straus S, et al. Primary care physicians' perspectives on computer-based health risk assessment tools for chronic diseases: a mixed methods study. J Innov Heal informatics . 2015;22(3):333–9.
47. Porat T, Delaney B, Kostopoulou, O. The impact of a diagnostic decision support system on the consultation: perceptions of GPs and patients. BMC Med Inform Decis Mak. 2017;17(1):79.
48. McParland C, Cooper M. Differential Diagnosis Decision Support Systems in Primary and Out-of-Hours Care: A Qualitative Analysis of the Needs of Key Stakeholders in Scotland. J Prim Care Community Health. 2019;10:2150132719829315.
49. Kukafka R, Yi H, Xiao T, et al. Why Breast Cancer Risk by the Numbers Is Not Enough: Evaluation of a Decision Aid in Multi-Ethnic, Low-Numerate Women. J Med Internet Res . 2015;17(7):e165. 93442
50. Merkebu, J. *et al.* Situativity: a family of social cognitive theories for understanding clinical reasoning and diagnostic error. Diagnosis 7, 2020, 169–176.
51. Singh H, Schiff GD, Graber ML, et al. The global burden of diagnostic errors in primary care. BMJ Qual Saf. 2016/08/16. 2017 Jun;26(6):484–94.
52. Elwyn G, Frosch D, Thomson R, et al. Shared decision making: a model for clinical practice. J Gen Intern Med. 2012 Oct 27(10):1361–7.
53. Berger ZD, Brito JP, Ospina NS, et al. Patient centred diagnosis: sharing diagnostic decisions with patients in clinical practice. BMJ, 2017 Nov 1; 359:j4218.
54. Hatch, S. Uncertainty in medicine. BMJ, 2017, 357:j2180
55. Hillen, MA, Gutheil, CM., Strout, TD, et al. Tolerance of uncertainty: Conceptual analysis, integrative model, and implications for healthcare. Social Science and Medicine, 2017. vol. 180 62–75
56. Cox, CL, Miller, BM, Kuhn, I, et al. Diagnostic uncertainty in primary care: what is known about its communication, and what are the associated ethical issues? Family Practice, 2021. doi:10.1093/fampra/cmab023.
57. Coma, E, Medina, M, Mendez, L et al. Effectiveness of electronic point-of-care reminders versus monthly feedback to improve adherence to 10 clinical recommendations in primary care: a cluster randomized clinical trial. BMC Med Inform Decis Mak*, 2019* **19**, 245.
58. Schedlbauer, A, Prasad, V, Mulvaney, C. et al. What Evidence Supports the Use of Computerized Alerts and Prompts to Improve Clinicians' Prescribing Behavior? J of the Am Med Inform Ass, 2009, 16, 531–538.
59. Jones C, Thornton J, Wyatt JC. Enhancing trust in clinical decision support systems: a framework for developers. BMJ Health and Care Informatics. 2021 Jun 1;28(1):e100247.
60. Spence, J. D. The need for clinical judgement in the application of evidence-based medicine.

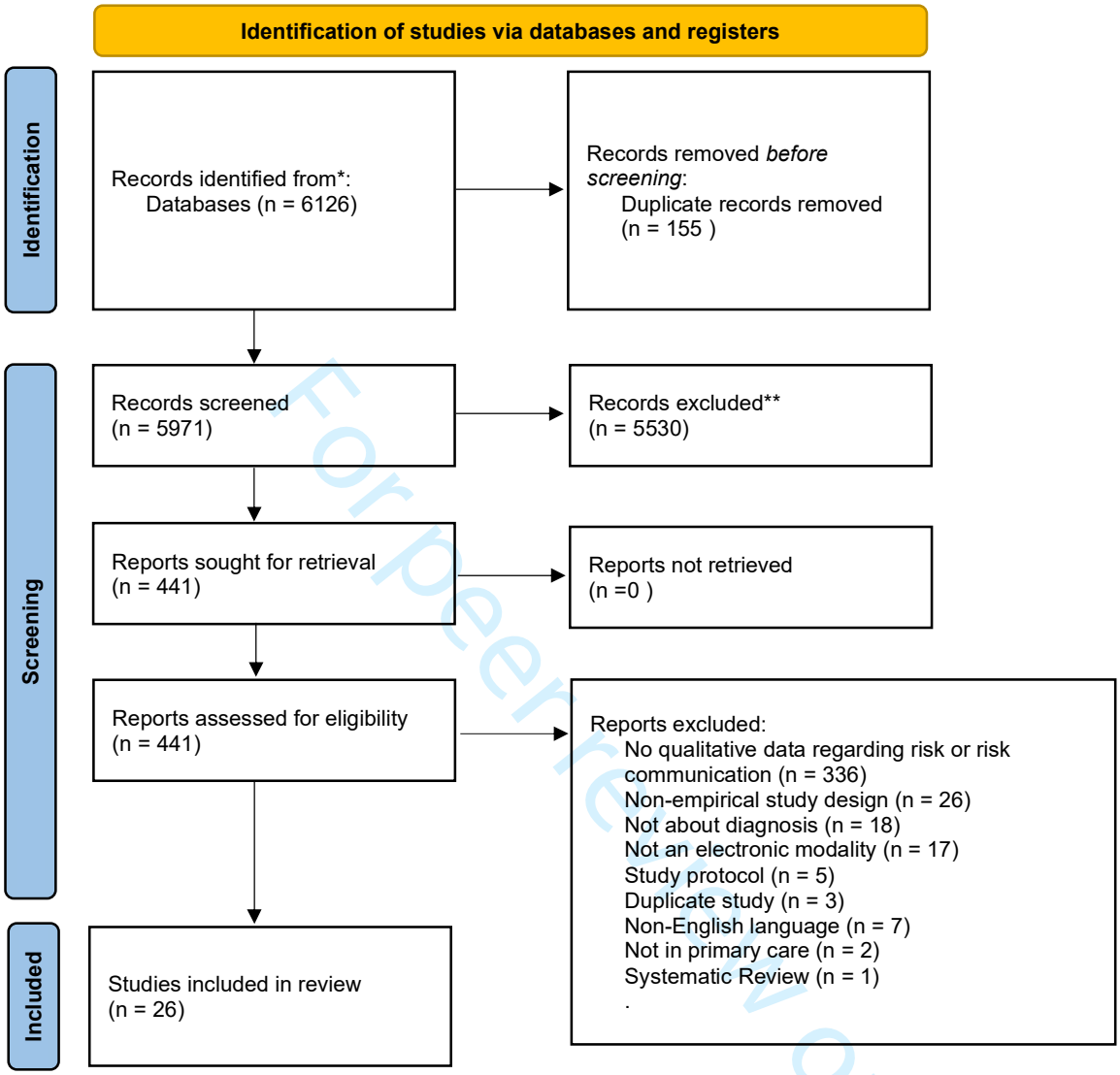
For peer review only

BMJ EBM 2020, 25, 172–177

61. Chartash, D., Sassoon, D. & Muthu, N. Physicians in the Era of Automation: The Case for Clinical Expertise. MDM policy & practice 2019, 4,
62. Friedemann Smith, C., Drew, S., Ziebland, S. & Nicholson, B. D. Understanding the role of GPs' gut feelings in diagnosing cancer in primary care: a systematic review and meta-analysis of existing evidence. Br J Gen Pract. 2020; 70, E612–E621
63. Hjertholm P, Moth G, Ingeman ML, et al. Predictive values of GPs' suspicion of serious disease: a population-based follow-up study. Br J Gen Pract. 2014 Jun 1;64(623):e346 LP-e353.
64. Heath, I. Role of fear in overdiagnosis and overtreatment—an essay by Iona Heath. BMJ, 2014; 349, g6123
65. Witte K, Allen M. A Meta-Analysis of Fear Appeals: Implications for Effective Public Health Campaigns. Heal Educ Behav. 2000 Oct 1;27(5):591–615.
66. Leventhal H. Findings and Theory in the Study of Fear Communications. In: Advances in Experimental Social Psychology, editor Berkowitz L. Academic Press; 1970. p. 119–86.
67. Campbell, J. L. *et al.* Policies and strategies to retain and support the return of experienced GPs in direct patient care: the ReGROUP mixed-methods study. Health Services and Delivery Research 2019, vol. 7, 14

For peer review only

Fig 1: Flow Diagram for systematic searches of databases and registers



For peer review only

Supplementary Table 1: Characteristics of Included Studies

Study Author and Date	Country of Subjects	Diagnosis of interest	eRAT form	Nomenclature used in study	Research Participant	eRAT exposure	Data Collection Methodology	Data Analysis	Risk/uncertainty theme in 1° analysis	CASP Quality Assessment
Babb de Villiers 2019	UK and international	Cancer	Prognostic	Risk prediction models	Clinicians	Workshop	Workshop	Thematic Analysis	No	**2**
Chiang 2015	Australia	Cancer	Clinical	Risk tool	Clinicians	Simulated Clinic	Semi-structured Interviews	NPT	No	**8**
Coe 2016	USA	Breast Cancer	Prognostic	Risk Assessment	Patients	Workshop/Usability testing	"Morae" software recording.	Framework Analysis (implied not stated)	No	**8**
Collins 2013	Australia	Breast cancer	Prognostic	Risk assessment and decision support	Clinicians	Workshop	Focus groups	Unclear	No	**4**
Colombet 2003	France	Cardiovascular Disease, Cancer, MH risks	Prognostic	Personalised Estimate of Risk	Clinicians	Workshop/Usability testing	Focus group	Framework Analysis (implied not stated)	Yes	**6**
Crawford 2010	UK	DM foot disease	Prognostic	Risk Assessment Tool	Clinicians	Real World Use	Semi-structured interviews	Thematic Analysis	No	**8**
Dikimitis 2015	UK	Cancer	Clinical	Electronic Risk Assessment Tools	Clinicians	Real World Use	Semi-structured interviews	NPT	No	**8**
Du 2010	China	Cardiovascular Disease	Prognostic	Risk Assessment Tools	Clinicians	Workshop/Usability testing	Semi Structured focus groups	Inductive coding	No	**3**

For peer review only

Eaton 2009	USA	Cardiovascular Disease	Prognostic	Interactive Risk Communication and Decision Support	Patients	Real World Use	Focus Groups	Unclear	Yes	**4**
Harry 2019	USA	Cancer Cardiovascular Disease	Prognostic	Clinical Decision Support	Clinicians	Real World Use	Semi-structured interviews	Grounded Theory	No	**9**
Hawking 2019	UK	Cardiovascular Disease	Prognostic	Risk Report	Patients	Real World Use	Semi-structured interviews	Thematic Analysis	Yes	**8**
Jiwa 2006	USA	Cancer	Clinical	Interactive referral proforma	Clinicians	Real World Use	Semi structured interviews	Thematic Analysis	No	**7**
Keogh 2019	Australia and New Zealand	Breast Cancer	Prognostic	Risk assessment and management support tool	Clinicians and Patients	Workshop/Usability Testing	Focus Group	Thematic Analysis (implied not stated)	No	**8**
Kidney 2017	UK	Colorectal Cancer	Clinical	Information based local evaluation	Clinicians and Practice Managers	Real World Use	Interviews	Thematic Analysis	Yes	**6**
Kukafka 2015	USA	Breast Cancer	Prognostic	Decision aid	Patients	Real World Use	Focus groups	Thematic Analysis (implied not stated)	Yes	**8**
McParland 2019	UK	DDx	DDx	Decision Support System	Clinicians	Real World Use	Focus groups	Thematic Analysis	No	**8**
Nieroda 2018	UK	Cancer	Clinical	Decision Support Tool	Clinicians and Patients	Workshop/Usability Testing	Focus groups, think aloud use of website, open	Thematic Analysis	No	**8**

For peer review only

							ended questionnaires			
Peiris 2009	Australia	CVD	Prognostic	Clinical Decision Support	Clinicians	Real World Use	Semi-structured interviews	Thematic Analysis	Yes	**8**
Porat 2017	UK	DDx	DDx	Decision Support System	Clinicians	Simulated Clinic	Semi-structured interviews	Thematic Analysis	No	**7**
Porat 2018	UK	Stroke	Prognostic	Decision Aid	Clinicians	Workshop/Usability Testing	Semi-structured interviews	Thematic Analysis	Yes	**3**
Porat 2019	UK	Stroke	Prognostic	Decision Aid	Clinicians, Patients, and other Stakeholders	Workshop/Usability testing	Focus groups, Interviews and usability evaluations.	Thematic Analysis	Yes	**5**
Richardson 2019	USA	Pharyngitis and Cough	Clinical	Decision Support Tools	Clinicians	Real World Use	Interview, video and audio of consultation	Constant Comparative Method	No	**8**
Saver 2014	USA	Diabetes complications	Prognostic	Outcome Model	Patients	Real World Use	Interviews	general inductive approach	Yes	**5**
Short 2003	UK	Stroke	Prognostic	Decision Support System	Clinicians	Simulated Clinic	Qualitative interview and questionnaire	Unclear	No	**7**
Short 2004	UK	Stroke	Prognostic	Decision Support System	Clinicians	Workshop/Usability testing	Interviews	Constant Comparative Method	Yes	**8**
Vail 2012	UK	Mental Health risk	Prognostic	Decision Support System	Clinicians	Workshop/Usability testing	Interviews, focus groups	Thematic Analysis	No	**8**

For peer review only

Voruganti 2015	Canada	Multiple	Prognostic	Risk Assessment Tools	Clinicians	Workshop/Usability testing	Focus Groups and usability testing	Grounded theory	No	**8**
Walker 2017	Australia	Colorectal Cancer	Prognostic	Risk Prediction Tool	Clinicians and Practice Managers	Simulated Clinic	Interviews, videoed simulated consultations analysed	Framework Analysis	No	**7**

Supplementary Table 2: Assessment of quality of studies with the CASP tool

Study ID	1. Was there a clear statement of the aims of the research?	2. Is a qualitative methodology appropriate?	3. Was the research design appropriate to address the aims of the research?	4. Was the recruitment strategy appropriate to the aims of the research?	5. Was the data collected in a way that addressed the research issue?	6. Has the relationship between researcher and participants been adequately considered?	7. Have ethical issues been taken into consideration?	8. Was the data analysis sufficiently rigorous?	9. Is there a clear statement of findings?	10. How valuable is the research?
Babb de Villiers 2019	Yes	Yes	No	No	Can't tell	No	No	No	No	No
Chiang 2015	Yes	Yes	Yes	Yes	Yes	Can't tell	Yes	Yes	Yes	Yes
Coe 2016	Yes	Yes	Can't tell	Yes	Yes	No	Yes	Yes	Yes	Yes
Collins 2013	Yes	Yes	Yes	Can't tell	Can't tell	Can't tell	Can't tell	Can't tell	Yes	Yes
Colombet 2003	Yes	Yes	Yes	Yes	Yes	Can't tell	Yes	Yes	Yes	Yes
Crawford 2010	Yes	Yes	Yes	Yes	Yes	Can't tell	Yes	Yes	Yes	Yes
Dikimitis 2015	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Du 2010	Yes	Yes	Yes	Can't tell	Can't tell	Can't tell	Can't tell	Can't tell	Can't tell	Can't tell
Eaton 2009	Yes	Yes	Can't tell	Can't tell	Yes	No	No	Can't tell	Yes	Yes
Harry 2019	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

For peer review only

Hawking 2019	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Jiwa 2006	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Can't tell	Can't tell
Keogh 2019	Yes	Yes	Yes	Yes	Yes	Can't tell	Yes	Yes	Yes	Yes
Kidney 2017	Yes	Yes	Yes	Can't tell	Yes	No	Yes	Can't tell	Yes	Yes
Kukafka 2015	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
McParland 2019	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Nieroda 2018	Yes	Yes	Yes	Yes	Yes	Can't tell	Yes	Yes	Yes	Yes
Peiris 2009	Yes	Yes	Yes	Yes	Yes	Can't tell	Yes	Yes	Yes	Yes
Porat 2017	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Porat 2018	No	Yes	Can't tell	Yes	Yes	No	No	No	Yes	Can't tell
Porat 2019	Yes	Yes	Yes	Yes	Yes	No	Can't tell	Can't tell	Yes	Can't tell
Richardson 2019	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Saver 2014	Yes	Yes	Can't tell	Yes	Can't tell	No	Yes	Yes	Yes	Can't tell
Short 2003	Yes	Yes	Yes	Can't tell	Yes	Can't tell	Yes	Yes	Yes	Yes
Short 2004	Yes	Yes	Can't tell	Yes	Yes	No	No	No	Yes	Yes
Vail 2012	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Voruganti 2015	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Walker 2017	Yes	Yes	Yes	Yes	Can't tell	Can't tell	Yes	Yes	Yes	Yes

Supplementary Table 3: Illustrative Quotes to reinforce Analytical Themes

Theme		Quote (all clinician comments, unless stated)	Study

For peer review only

Novel Risk	*Alerts to new risk*	*"It's just electronic highlight, makes you think even if you immediately dismiss it, at least that millisecond you've thought about it, and I think that is going to be useful at some point, but for how many people I don't know"*	Dikomitis 2015
		"Helpful to refresh your memory about things that are rare…we mainly stick to the phrase: 'common things are common"	Porat 2017
	Alerts to alternative risks	*"Yes, I must admit ovarian didn't come so high up, I had already fixated on the bowel stuff, which maybe I shouldn't have. This really said hey, consider ovarian as well. So again that was useful, yes"*	Chiang 2015
		"I think it widens your diagnosis, like the guy that had COPD and then ended up with AS. I would get to the diagnosis, but it would be longer without the tool."	Porat 2017
	Alerts to risk factor	*"the patient can see very clearly, [and say] 'Wow, smoking contributes a lot to cardiovascular risk, I do actually want to think about smoking cessation'"*	Voruganti 2015
		"I don't think it's (the eRAT) altered practice. I think it's kept some of the not so hard red flag symptoms in people's minds…"	Kidney 2017
Risk Refinement	*Orientation of risk (within guidelines/clinical pathways)*	*"Normally I'd get a few investigations, get the results back and then based on that say do we need to do something, or I refer this on based on that. But I guess if I have a calculator saying it's higher risk, it might prompt me to make a referral to a specialist a bit earlier."*	Chiang 2015
		"Think the biggest use I got, because you presented with very constitutional symptoms that a lot of things could present with, this was useful for me to think what's the good next step and what should I think of. So I was thinking in my mind more"	Chiang 2015
	More accurate risk, reduction of uncertainty	*For clinicians, the main perceived benefit was iPrevent's provision of accurate and reliable risk estimates for both ten year and lifetime risk, and standardized risk management advice for women*	Keogh 2019, Researcher comment
		"It is helpful in that it reminds me that the risk of stroke goes up substantially as you get older so your benefit (of aspirin) increases"	Short 2003
	Simplification of risk, reduction of uncertainty	*"Well I suppose you could think about the common comorbidities, so hypertension and stroke, AF (atrial fibrillation) and stroke, diabetes and stroke and you could think about not necessarily an algorithm but a sort of stepwise prioritisation about what you should think about in terms of the patient's total management, you know, which would be the most important area of focus?"*	Porat 2019

For peer review only

	Introduction of new risk knowledge	*"Other colleagues have said as well we're far more aware of thrombocytosis and increased platelets. We weren't aware that tended to increase the score for increased cancer risk"*	Dikomitis 2015
		"The learning aspect is very important, since this system is based on evidence and evidence can change"	Porat 2019
Communication	*Patient to Clinician Communication*	*"I think it's important to be informed and to get there as early as possible. To go and see the GP, and if there's a tool perhaps, you know, to help me do this, that may be of interest"*	Nieroda 2018 Patient Comment
	Risk focuses conversations	*"it usually stops a lot of the meandering dialogue that you'd otherwise engage in…. [initiate] that discussion in terms of what their risk factors are"*	Voruganti 2015
		"So having that piece of paper there gave you that conversation: 'Well now we've finished everything, let's look at this.'"	Peiris 2009
	Risk justifies action	*"But your heart is beating kinda fast, you've had a fever last night...the recommendation would be to get a chest x-ray today."*	Richardson 2019
		"OK, so our little risk calculator here is recommending that we would swab you for strep throat, and I agree with that."	Richardson 2019
	Risk justifies inaction	*"…there's a whole lot of people and this is your chance out of them of cancer…"*	Walker 2007
		"I think FRAX [is] useful because [it] can cut down on unnecessary investigations"	Voruganti 2015
	Improves understanding of risk	*"I think the biggest impact is that it changed the way I talked about what I was doing with patients, in that it made it a much more slick, neat package to describe the normal screening that you do for risk management. And so I felt it was easier to deliver some description of where they're at now. And from their point of view, I mean it's hard to know, but they seemed to understand that it was a multifactorial thing, rather than just being one of those single disease problems.... The thing that I don't really know, that I guess would be useful, is what they think when they walk out the door, what they actually understand of what I've said."*	Pieris 2009
		"With our Framingham tool you get…, a yellow bar. The average person's 10-year risk is…two little yellow bars, and your [the patient's] average risk is…maybe five little yellow bars'"	Voruganti 2015

For peer review only

		"I like this one [referring to the color bar].... I mean, everyone knows that red means danger, so if they're heading towards this one, it's a lot more visual, the impact...."	Peiris 2009
Autonomy	*Empowered by knowing personalised risk*	*"Anything that can be personalised or tailored, so you don't feel it's this off the shelf thing that you're being given, you know… you sit with your doctor and it's not just a case of giving out a leaflet, but actually let's have a look at your personal data"*	Porat 2019
		"I think the benefit is you're empowering yourself with information, you're possibly changing what you can change"	Keogh 2019, Patient comment
		"Anything that gets people to have a think about their health is a good thing, anything at all."	Walker 2007
Fear	*Direct fear of diagnosis*	*"Well I just kind of—a heart attack is a scariest thing to me. And I think it just kind of got me wondering and thinking and got me too nervous about the whole thing"*	Eaton 2009, Patient comment
		"And I thought, 'Wow! Patients coming with these tummy pains,' even if I'm thinking, 'She's got a 10% chance,' to actually think, 'Wow! Listen, you got a 38% chance of cancer.' General public are very—I would find that confronting and the general public have a shocking record at understanding risk."	Chiang 2015
	Disturbs normal practice	*"It's completely different to how I normally work, unsettling and confusing, I'm not quite sure where I am, to what extent I'm following a template and how I take history."*	Porat 2017
		"I personally don't think tools help me too much, interfere with my process of thought, would have done better without it, it was unnatural"	Porat 2017
	Removes focus and control of consultation	*"it might bring up a lot more other issues that they [patients] weren't originally aware of and the discussion might actually… be less directed"*	Voruganti 2015
		'Yeah. If I start saying to them, 'Oh god, your risk of having gastro-oesophageal cancer is one in three'y that seems way over the top, back pedalling from that, the consultation would've gone on for another 20 minutes or 30 minutes since I tried to explain to him, no. He doesn't have it'	Chiang 2015
	Lack of agency over risk	*The most commonly expressed concern was the potential for the risk assessment tool to miss something that was unknown and outside their control*	Kukafka 2015, Researcher Comment
	Medico-legal fear	*"Quite a few partners were worried about any medico-legal implications with*	Dikomitis 2015

For peer review only

		that. I think the one criticism was having the list of patient with PPVs. No one really liked that. That was worrying. We're worried about if patients knew that you had a list of them with the risk and you hadn't acted on it, what would be the implications? That was probably a point that put people off, really"	
		Some also felt that overriding a DDDSS would make them vulnerable to litigation	McParland 2019, Researcher Comment
Mistrust	*eRAT over-triggers*	*"Inevitably it will come up with a diagnosis of cancer. When the patient walks through the door . . . quite clearly, quickly, we can see this is not a serious diagnosis"*	McParland 2019
		"It gives information which, as it's blandly presented, you go, "How did you get that?…" I got a couple of people where I got a 20% number and you go, "Oh that's madness, that's not you," and often because it's based on single digit information…like a single blood pressure."	Peiris 2009
	eRAT misses things	*A consistent theme in our analysis was uncertainty among participants that all factors were accounted for in determining their breast cancer risk*	Kukafka 2015, Researcher comment
	eRAT not applicable to their patient	*In half of the patient visits, patients reported some piece of information, typically as a part of the history of present illness that raised a question for the coders of whether the tool was applicable to their clinical condition.*	Richardson 2019, Researcher comment
		they felt obliged to use the tool for the purpose of the study and their suspicion of cancer was unnaturally heightened	Chiang 2015
	Preference for diagnostic tests over risk scores	*"I would like to know if I have the gene mutation. So for me to say that being my mom had it [breast cancer] and take a pill and prevent it and not knowing if I have the gene doesn't make sense to me."*	Kukafka 2015
		"Colonoscopy was commonly referred to as the 'gold standard' for CRC screening. GPs were more likely to recommend colonoscopy if the patient wanted one or they had previously been recommended one by a gastroenterologist, even if the patient was at average risk."	Walker 2007, Researcher comment
	Preference for clinical judgement	*"… more experienced GPs were less likely to agree with the underlying concept and continued to maintain strong belief in their clinical intuition."*	Chiang 2015, Researcher comment
		"Patients don't follow flow charts, in my experience"	McParland 2019

For peer review only

		…that's just the tricky bit of saying [to the patient] that this is what's [sic] the recommendation but we don't always follow what recommendations are…	Walker 2007
	Patient preference for self knowledge	*"I think the model is incorrect because I know I am going to last longer than 10 years, because I do take care of myself"*	Saver 2014

For peer review only

ENTREQ checklist (Enhancing transparency in reporting the synthesis of qualitative research) *

No. Item	Guide questions/description	Reported on Page #
1. Aim	State the research question the synthesis addresses	Title & 3
2. Synthesis methodology	Identify the synthesis methodology or theoretical framework which underpins the synthesis, and describe the rationale for choice of methodology (e.g. meta-ethnography, thematic synthesis, critical interpretive synthesis, grounded theory synthesis, realist synthesis, meta-aggregation, meta-study, framework synthesis)	4
3. Approach to searching	Indicate whether the search was pre-planned (comprehensive search strategies to seek all available studies) or iterative (to seek all available concepts until they theoretical saturation is achieved)	4
4. Inclusion criteria	Specify the inclusion/exclusion criteria (e.g. in terms of population, language, year limits, type of publication, study type)	4
5. Data sources	Describe the information sources used (e.g. electronic databases (MEDLINE, EMBASE, CINAHL, psycINFO), grey literature databases (digital thesis, policy reports), relevant organisational websites, experts, information specialists, generic web searches (Google Scholar) hand searching, reference lists) and when the searches conducted; provide the rationale for using the data sources	4
6. Electronic Search strategy	Describe the literature search (e.g. provide electronic search strategies with population terms, clinical or health topic terms, experiential or social phenomena related terms, filters for qualitative research, and search limits)	Appendix
7. Study screening methods	Describe the process of study screening and sifting (e.g. title, abstract and full text review, number of independent reviewers who screened studies)	4
8. Study characteristics	Present the characteristics of the included studies (e.g. year of publication, country, population, number of participants, data collection, methodology, analysis, research questions)	Table 1
9. Study selection results	Identify the number of studies screened and provide reasons for study exclusion (e,g, for comprehensive searching, provide numbers of studies screened and reasons for exclusion indicated in a figure/flowchart; for iterative searching describe reasons for study exclusion and inclusion based on modifications to the research question and/or contribution to theory development)	4 and Figure 1
10. Rationale for appraisal	Describe the rationale and approach used to appraise the included studies or selected findings (e.g. assessment of conduct (validity and robustness), assessment of reporting (transparency), assessment of content and utility of the findings)	4

For Peer Review Only

11. Appraisal items	State the tools, frameworks and criteria used to appraise the studies or selected findings (e.g. Existing tools: CASP, QARI, COREQ, Mays and Pope [25]; reviewer developed tools; describe the domains assessed: research team, study design, data analysis and interpretations, reporting)	4
12.Appraisal process	Indicate whether the appraisal was conducted independently by more than one reviewer and if consensus was required	4
13.Appraisal results	Present results of the quality assessment and indicate which articles, if any, were weighted/excluded based on the assessment and give the rationale	Table 2
14. Data extraction	Indicate which sections of the primary studies were analysed and how were the data extracted from the primary studies? (e.g. all text under the headings "results /conclusions" were extracted electronically and entered into a computer software)	4
15. Software	State the computer software used, if any	4
16. Number of reviewers	Identify who was involved in coding and analysis	4
17. Coding	Describe the process for coding of data (e.g. line by line coding to search for concepts)	4
18. Study comparison	Describe how were comparisons made within and across studies (e.g. subsequent studies were coded into pre-existing concepts, and new concepts were created when deemed necessary)	4
19. Derivation of themes	Explain whether the process of deriving the themes or constructs was inductive or deductive	4
20. Quotations	Provide quotations from the primary studies to illustrate themes/constructs, and identify whether the quotations were participant quotations of the author's interpretation	Table 3
21. Synthesis output	Present rich, compelling and useful results that go beyond a summary of the primary studies (e.g. new interpretation, models of evidence, conceptual models, analytical framework, development of a new theory or construct)	5-7

* Reference: Tong A, Flemming K, McInnes E, Oliver SA, Craig J. Enhancing transparency in reporting the synthesis of qualitative research: ENTREQ. BMC Medical Research Methodology 2012, 12:181.

Search Strategy

#	Database	Search term	Results
1	Medline	("electronic risk assessment*").ti,ab	13
2	Medline	(electronic ADJ3 ("risk assessment*" OR RAT OR eRAT OR RATs OR eRATs OR diagnosis)).ti,ab	459

For peer review only

3	Medline	(comput* ADJ3 ("risk assessment*" OR RAT OR eRAT OR RATs OR eRATs OR diagnosis)).ti,ab	8879
4	Medline	"DECISION SUPPORT TECHNIQUES"/	19374
5	Medline	"DECISION SUPPORT SYSTEMS, CLINICAL"/ OR "DIAGNOSIS, COMPUTER-ASSISTED"/	29078
6	Medline	"RISK ASSESSMENT"/	249398
7	Medline	(electronic OR comput*).ti,ab	805823
8	Medline	(6 AND 7)	10675
9	Medline	("decision support").ti,ab	13295
10	Medline	("prediction tool*").ti,ab	3009
11	Medline	(qcancer).ti,ab	10
12	Medline	("risk prediction").ti,ab	8010
13	Medline	("computer guided consultation").ti,ab	2
24	Medline	(1 OR 2 OR 3 OR 4 OR 5 OR 8 OR 9 OR 10 OR 11 OR 12 OR 13)	82373
14	Medline	exp "GENERAL PRACTICE"/	73872
15	Medline	"PRIMARY HEALTH CARE"/	73929
16	Medline	"GENERAL PRACTITIONERS"/	7252
17	Medline	"PHYSICIANS, FAMILY"/	16158
18	Medline	("general practi*").ti,ab	38963
19	Medline	("primary health care").ti,ab	19769
20	Medline	("primary care").ti,ab	106822
21	Medline	(GP OR GPs).ti,ab	55107

For peer review only

22	Medline	("family physician*").ti,ab	14422
23	Medline	("family practice").ti,ab	7465
25	Medline	(14 OR 15 OR 16 OR 17 OR 18 OR 19 OR 20 OR 21 OR 22 OR 23)	274818
26	Medline	(24 AND 25)	3066
27	EMBASE	("electronic risk assessment*").ti,ab	30
28	EMBASE	(electronic ADJ4 ("risk assessment*" OR RAT OR eRAT OR RATs OR eRATs OR diagnosis)).ti,ab	811
29	EMBASE	(comput* ADJ4 ("risk assessment*" OR RAT OR eRAT OR RATs OR eRATs OR diagnosis)).ti,ab	12523
30	EMBASE	exp "DECISION SUPPORT SYSTEM"/ OR "COMPUTER ASSISTED DIAGNOSIS"/	61278
31	EMBASE	"RISK ASSESSMENT"/	534341
32	EMBASE	(electronic OR comput*).ti,ab	1181113
33	EMBASE	(31 AND 32)	24951
34	EMBASE	("decision support").ti,ab	17689
35	EMBASE	("prediction tool*").ti,ab	4544
36	EMBASE	(qcancer).ti,ab	14
37	EMBASE	("risk prediction").ti,ab	12919
38	EMBASE	("computer guided consultation").ti,ab	8
39	EMBASE	(27 OR 28 OR 29 OR 30 OR 33 OR 34 OR 35 OR 36 OR 37 OR 38)	117361
40	EMBASE	"GENERAL PRACTICE"/	76245
41	EMBASE	exp "PRIMARY HEALTH CARE"/	162467

For peer review only

42	EMBASE	"GENERAL PRACTITIONER"/	94915
43	EMBASE	("general practi*").ti,ab	96563
44	EMBASE	("primary health care").ti,ab	23923
45	EMBASE	("primary care").ti,ab	146773
46	EMBASE	(GP OR GPs).ti,ab	79393
47	EMBASE	("family physician*").ti,ab	17848
48	EMBASE	("family practice").ti,ab	7988
49	EMBASE	(40 OR 41 OR 42 OR 43 OR 44 OR 45 OR 46 OR 47 OR 48)	415195
50	EMBASE	(39 AND 49)	4595
51	CINAHL	("electronic risk assessment*").ti,ab	7
52	CINAHL	(electronic ADJ4 ("risk assessment*" OR RAT OR eRAT OR RATs OR eRATs OR diagnosis)).ti,ab	279
53	CINAHL	(comput* ADJ4 ("risk assessment*" OR RAT OR eRAT OR RATs OR eRATs OR diagnosis)).ti,ab	2066
54	CINAHL	"DECISION SUPPORT SYSTEMS, CLINICAL"/	4554
55	CINAHL	exp "DECISION SUPPORT TECHNIQUES"/	9645
56	CINAHL	"DIAGNOSIS, COMPUTER ASSISTED"/	4031
57	CINAHL	"RISK ASSESSMENT"/	87369
58	CINAHL	(electronic OR comput*).ti,ab	184706
59	CINAHL	(57 AND 58)	4195
60	CINAHL	("decision support").ti,ab	5671
61	CINAHL	("prediction tool*").ti,ab	678

For peer review only

62	CINAHL	(qcancer).ti,ab	8
63	CINAHL	("risk prediction").ti,ab	2809
64	CINAHL	("computer guided consultation").ti,ab	1
65	CINAHL	(51 OR 52 OR 53 OR 54 OR 55 OR 56 OR 59 OR 60 OR 61 OR 62 OR 63 OR 64)	29549
66	CINAHL	"FAMILY PRACTICE"/	22717
67	CINAHL	"PHYSICIANS, FAMILY"/	17517
68	CINAHL	"PRIMARY HEALTH CARE"/	56524
69	CINAHL	("general practi*").ti,ab	26608
70	CINAHL	("primary health care").ti,ab	10032
71	CINAHL	(GP OR GPs).ti,ab	18016
72	CINAHL	("family physician*").ti,ab	5248
73	CINAHL	("family practice").ti,ab	1902
74	CINAHL	(66 OR 67 OR 68 OR 69 OR 70 OR 71 OR 72 OR 73)	111915
75	CINAHL	(65 AND 74)	1406

Web of Science Search

(“electronic risk assessment” OR electronic NEAR/3 ("risk assessment*" OR RAT OR eRAT OR RATs OR eRATs OR diagnosis) OR comput* NEAR/3 ("risk assessment*" OR RAT OR eRAT OR RATs OR eRATs OR diagnosis) OR “decision support” OR "prediction tool*" OR qcancer OR “risk prediction” OR "computer guided consultation") AND (“General practi*” OR “Primary Health Care” OR “Family Physician*” OR “Primary Care” OR GP OR GPs OR “Family Practice”)

For peer review only